

## L3 Digital Technologies Scholarship Project

Github: [https://github.com/Routund/School\\_Search](https://github.com/Routund/School_Search)

### INQUIRY MILESTONES

| What                                | Time spent | Milestone | Completed<br>✓ |
|-------------------------------------|------------|-----------|----------------|
| Establish focus & questions         | 1 hour     |           | ✓              |
| Undertake research                  | 5 hours    |           | ✓              |
| Organise, refine & analyse research | 2 hours    |           | ✓              |
| Write proposal                      | 3 hours    |           | ✓              |
| Summarise my findings               |            |           |                |

### INQUIRY

#### Inquiry Focus

File storage and retrieval

#### Big Question

What is the best way to search and store information for 10's of thousands files for burnside high school

#### Initial questions to help answer my big question?

- What are the common ways big companies use to search through their files and/or the internet, eg. Documentation providers, Wikipedia, Google?
- What's the best SQL database to be storing the data for a file system in
- How do search indexers find the context of words?

#### Inquiry Research

***What are the common ways big companies use to search through their files and/or the internet, eg. Documentation providers, Wikipedia, Google?***

Source #1: [Cornell University - Why is Google Search so fast?](#) - Cornell University - 16/10/16

- Google uses a technique called search indexing to search documents
- This involves making a database that stores each unique word in all the documents, and all the documents that the word appears in, something like:

|        |                                    |
|--------|------------------------------------|
| 'Dog'  | Document 1, Document 5, Document 6 |
| 'Cat'  | Document 1, Document 3, Document 4 |
| 'Care' | Document 2, Document 4             |
| 'Toy'  | Document 5                         |
| 'Vet'  | Document 2, Document 3, Document 6 |

- They then look at each word in a query, and only search through the documents that contain the word by looking up the word in the database, kind of like a index in an encyclopedia

**Source #2: [A simple guide to AI search](#) - Algolia - 23/7/24**

- Many Search engine companies are considering moving over to AI Search for their search engines (eg. Bing and Google)
- AI Search vectorises documents (turning them into positions in a space with thousands of dimensions instead of the normal 3) based off their context
- It also vectorizes the prompt, and using these vectors, finds the documents that are close in context to the prompt. This whole process is called Retrieval Augmented Generation (RAG)
- It then feeds the relevant documents into a Large language model along with the prompt so that the LLM knows context that is unique to the database.

**Summary**

The 2 current ways to do searching are search indexing and AI searches. While searching seems to have been long done by search engines using search indexing, AI is a technology many are considering to switch to for their search engines. The relative age of these 2 technologies can be seen through the age of articles about them. Search indexing articles often date back 8-10 years; everyone getting more connected to the internet and becoming curious about how search engines work. Meanwhile, articles about AI search and RAG are all from around 1-2 years ago. Curiously, a lot of the articles for AI Search are from AI companies (like Algolia, the source for AI search being a company specializing in AI searches for other companies), which could mean that there's an interest from AI companies to push the merits of AI search. It would be prudent to talk to my stakeholders (the school) to see what kind of search to use.

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***What is the best SQL Database to store the information for a file system?***

**Source #1: [PostgreSQL vs MySQL: The Definitive Comparison](#) - Antonello Zanini**

- Both PostgreSQL and MySQL have capabilities to do a full text search on text in a database. However, PostgreSQL seems to be more feature rich in its implementation. It is important to note that it would have to search text, not pdf documents.
- Postgres has concurrent read and write support, while MySQL does not. That means that multiple operations can't be made on the same database in MySQL, while they can in PostgreSQL.
- The author states that Postgres is better in write heavy scenarios (editing lots of data), while MySQL is better in read heavy scenarios (reading lots of data).

**Source #2: [SQLite: what are the practical limits?](#) - Stack Overflow**

- This is a question on popular IT help forum Stack Overflow, in it, a user is asking what limits does SQLite have in terms of how fast it can query large amounts of data
- Updates from the original poster and answers talk about how querying the database is mostly slowed down by going through too many tables in one query.

**Source #3: [Kinds of databases](#) - PythonAnywhere**

- If I were to use a server based database like MySQL or Postgres, I would need to host the server somewhere when giving it in for marking for scholarship.
- The free python hosting service PythonAnywhere allows MySQL servers for free, and SQLite can be stored as a file and so wouldn't need hosting. However, PostgreSQL requires a subscription

**Summary:**

PostgreSQL would be a good choice for the SQL Server if I was in industry, due to it's good full text search and it's array typing making it good for making reverse indexes, and it supporting the tsvectors data type means it could also support a vector database for an AI search. On the other hand, MySQL is better suited for reading the raw data faster, and also has many of the same features as PostgreSQL, if not to the same calibre. SQLite and its file based structure would be the easiest to set up, is fastest in reading due to its lightweight structure, but doesn't support a native fulltext search.

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***How do Search Indexers detect if two words are the same word with different suffixes/prefixes?***

**Source #1: [What is stemming?](#) - IBM**

- Search tokenizers use a process called stemming to turn words that have the same kind of root word at the start into the same word, eg: "Dancer", "Dances", "Dancing", "Dance" -> "Danc"
- There are multiple algorithms to do it, but the most up to date one seems to be Snowball stemming and Lancaster stemming.
- There are python libraries that can do stemming (NLTK)

**Source #2: [Stemming and Lemmatization](#) - Stanford Natural language processing group**

- Lemmatization works by effectively looking up each word in the dictionary to see what its root word is, or its *lemma*.
- This is much slower than stemming, but more accurate.

- Both Stemming and Lemmatization can't be perfectly accurate at finding associated words, due to things like context based differences for words (eg. Cardinal could mean a priest, or the bird)
- The python NLTK library can also do lemmatization.

### Summary

Stemming and Lemmatization seem to be the main ways to distill a word down to it's context. Search engines would use stemming because it's faster and the internet is just so big. If I am to choose between the two, it would be good to get a rough estimate for how many files there are that need to be indexed. The sources are two reputable places. IBM, one of the oldest computer science adjacent companies, and a research group from Stanford University, one of the leading universities in computer science, so I have valid reasons to trust the information in these articles.

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### Analysis

My inquiry was about how best to store, search, and retrieve files. This led me to asking, what's the best database to store files and what's the best way to search through files (mainly text based files like pdfs).

For the longest time, file search seems to have been done with a technique called search indexing. This involved storing a list of documents that contain each unique word. This technique is used by search engine companies scanning the web, as well as internal search engines for sites like wikipedia or the nytimes. Recently, big tech companies have been moving their own internal documentation into AI powered searches, using a technique called Retrieval Augmented Generation. This involves using vectors to find documents relative to a query, and feeding those documents into a Large Language model to give an answer relative to the prompt and context.

Out of these two, search indexing seems to be better for now. It has more documentation on how to do it because of its relative age, it has less societally negative bias, and it's more secure. I should be using search indexing for my project

The major question when it comes to a file system is how you're storing the file's search context. While the files themselves would be stored in folders, or references to them in external cloud/third party databases, the metadata about what a file's context would need to be stored separately. This context storage isn't vanishingly small, so it's a valid concern. SQL Databases are the standard way to store data in any type of application, however, there are many SQL frameworks. I've chosen to research 3 Open source SQL frameworks, PostgreSQL, MySQL, and SQLite.

I'm apprehensive to use PostgreSQL, because while it has support for array types and has concurrency support, it is server based. This means it's harder to test and deploy, especially at a small scale like mine. The other thing is that most services to host PostgreSQL servers (which I would need if I wanted the project to be used outside of my own computer) are paid, even if they give free hosting for other types of SQL Servers. Thus, I don't want to use PostgreSQL.

MySQL is a bit better of a choice, because it's better at handling read heavy contexts, like a search engine. It's also much easier to find free hosting for it.

SQLite is the last option. It is file based and lightweight, so I could package a database file with my project, and it would work out of the box. It's also the fastest, because of how lightweight it

is. The only drawback is it's very bad at doing complicated SQL queries. For a single person project like mine, with no money and a smaller user base than an industrial application, SQLite should be fine. It's also important for me that the school be able to maintain the database, and the school has plenty of experience with SQLite.

When search indexes scan through documents to create their index, they try to find the root context for words. The two main ways are Stemming and Lemmatization. Stemming involves cutting off the ends and starts of words to try and get the root word. Lemmatization involves looking up a given word in its own dictionary of context for each word.

An advantage of using these 2 methods is that there are free libraries in Python (My Programming language of choice) to automatically do Lemmatization and Stemming.

Lemmatization is more accurate, while Stemming is faster. Both however, cannot glean the context of a word based off the words around it. Whether I use one or the other depends on how many files the school has. If more, use stemming. If less, Lemmatization. This process only needs to be done once for each file.

The best way to store and search files for my context is a search index running an SQLite Database (Not industry standard, but better for a small project like mine), and tokenizing words using a Lemmatizer or Stemmer.

## PROPOSAL

### Context

At the start of the year, the head of department came into our class to talk about this very scholarship. In it, he mentioned that he wanted someone to make an AI model to process all the files in various storage providers like Schoology and Kamar. I initially thought that an AI model was overkill, surely a simple search would be better? He explained that since there were so many files, performing a simple search on them would take way too long, and wouldn't be able to search through the content of the documents.

I decided to take up the project, and started researching how to process documents so that an AI would have access to it. After talking to more teachers, classmates, and mulling over the problem, I had the epiphany that I could make a search engine like Google to search through the files.

After a while, I talked to the principal about the project. The main problem from him was that new teachers and staff were coming in, and it was hard to onboard the tons of information about safety procedures, what forms to fill out for what purpose, how to call in sick, etc. etc.

### Problem and/or Issue

It has become harder for teachers to search for files they need across all these different storage mediums, especially since the search provided for them is quite slow. This leads to teachers having to waste time trying to pull up files for the class. Students also have trouble finding resources like past papers across all the different storages. This is taking valuable time away from actually teaching.

### Scope

I want this search engine to be used by students and teachers to search through resources that teachers have uploaded to Schoology, and for only teachers to search resources uploaded to KAMAR and our school's proprietary drives.

## Purpose

Provide a way for users to search Burnside High Schools files to find course content for their course, whether that be teacher or student.

## Requirements & Specifications

| Requirement                                          | Specifications                                                                                                                                                           | Relevant Implication           |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Be fast                                              | <ul style="list-style-type: none"><li>• Use a search index to quickly search through the documents</li><li>• Use the minimum amount of SQL Queries as possible</li></ul> | Usability                      |
| Be secure                                            | <ul style="list-style-type: none"><li>• Use gmail authentication to handle accounts</li><li>• Only store links to each document, not the content</li></ul>               | Security                       |
| Be easily maintainable by staff after I leave school | <ul style="list-style-type: none"><li>• Use Python and SQLite</li><li>• Have a non tech user friendly admin view</li></ul>                                               | Future Proofing/Sustainability |

## End Users

- Teachers at Burnside High School
- Students (Maybe)?
- Parents (Also a maybe)?

## Resources

- Access to school files
- Open source SQL Database frameworks
- Python (Programming Language)

## RISKS

### Security:

A finished search engine would have, at the minimum, gone through the text of all the files in the schools systems. This could include files that not all users should have access to. If the permissions are not properly preserved, users would be able to gain access to files they can't. This is not the biggest issue if the only users are teachers. However, if it was open to students while the permissions were not rock solid, there could be issues with students accessing assessments and tests before they should have access to them.

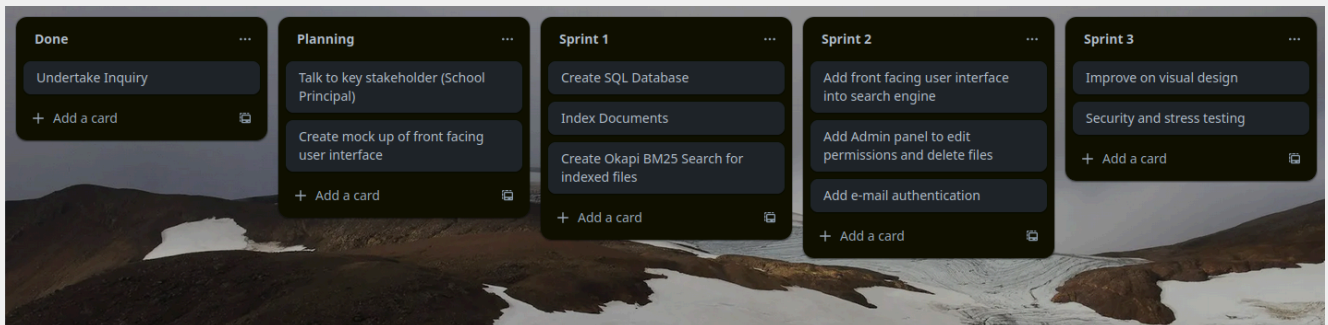
## Space and memory:

Indexing all of the documents could take up too much space in the school's servers if it's on site, or surpass the storage limit on a cloud server the school has. Compressing the database is going to be key to making sure that I have relatively small file sizes.

## INQUIRY SUMMARY

There are many ways to search, but the two main ways are search indexing and AI Search. For my purposes a search index is better due being more reliable, being cheaper, and more reliably friendly

## PROJECT MANAGEMENT



## DESIGN

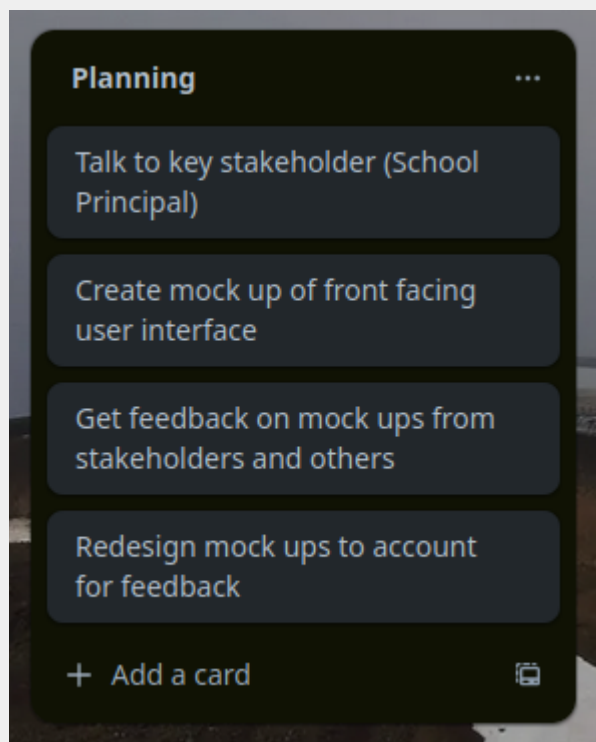
### Purpose

Provide a way for users to search Burnside High Schools files to find course content for their course, whether that be teacher or student.

### End User Requirements

- Be Fast
- Be Secure
- Be Maintainable

## Design Planning



## Relevant Implications

| Implication                        | Explain the implication                                                                         |
|------------------------------------|-------------------------------------------------------------------------------------------------|
| Social                             | Is the project socially appropriate                                                             |
| Cultural                           | Is the project culturally sensitive                                                             |
| Legal                              | Is the project legal to make                                                                    |
| Ethical                            | Is the project morally acceptable                                                               |
| Intellectual Property              | Does the project have permission to use everything that is contained in the website/            |
| Privacy                            | Does the project handle user information safely and securely                                    |
| Accessibility                      | Is the project usable by those may have impairments in using the project                        |
| Usability                          | Is the project easy to use                                                                      |
| Functionality                      | Does the project do what it's supposed to do                                                    |
| Aesthetics                         | Is the project nice for the senses                                                              |
| Sustainability and Future Proofing | Does the project work well into the future, and with how much applied effort to keep it working |



|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| End-user considerations | Does the project meet the needs of end users            |
| Health and Safety       | Does the project pose a health or safety risk to anyone |

Research

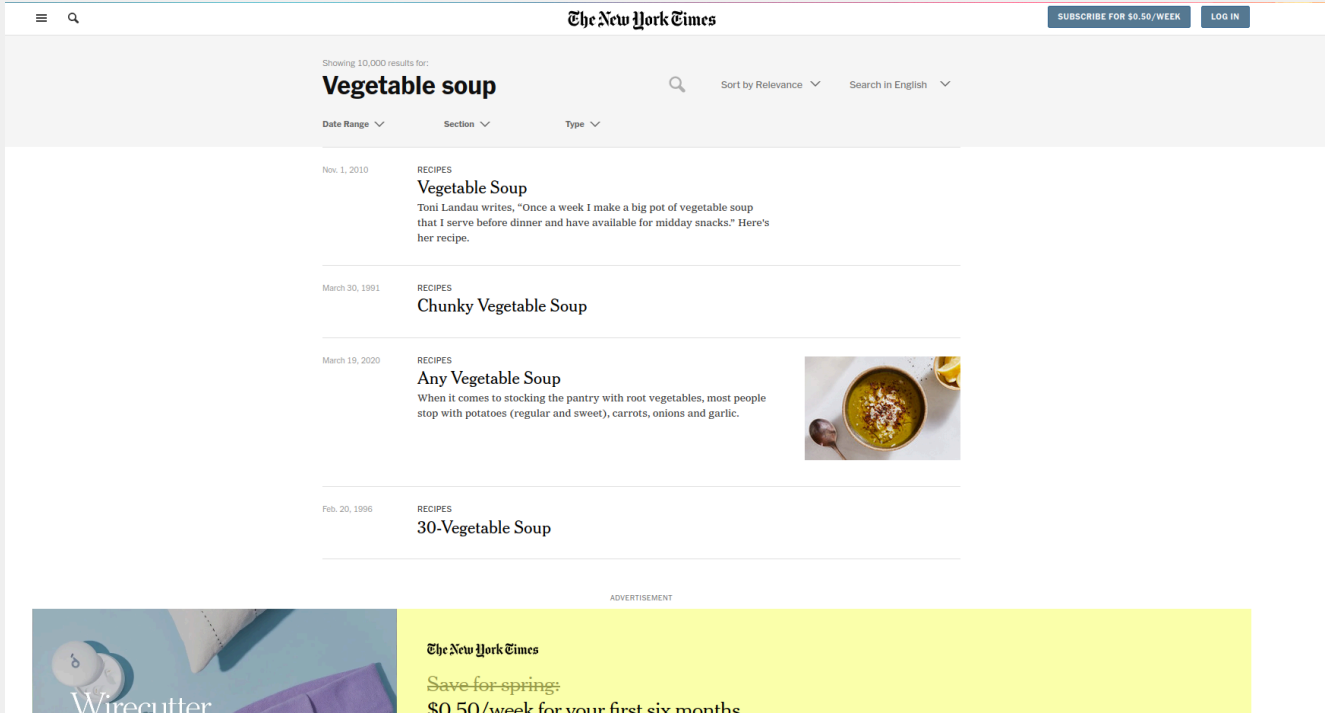
RELEVANT UX METHODOLOGIES

QUBStudio - Best UX practices for search interface -

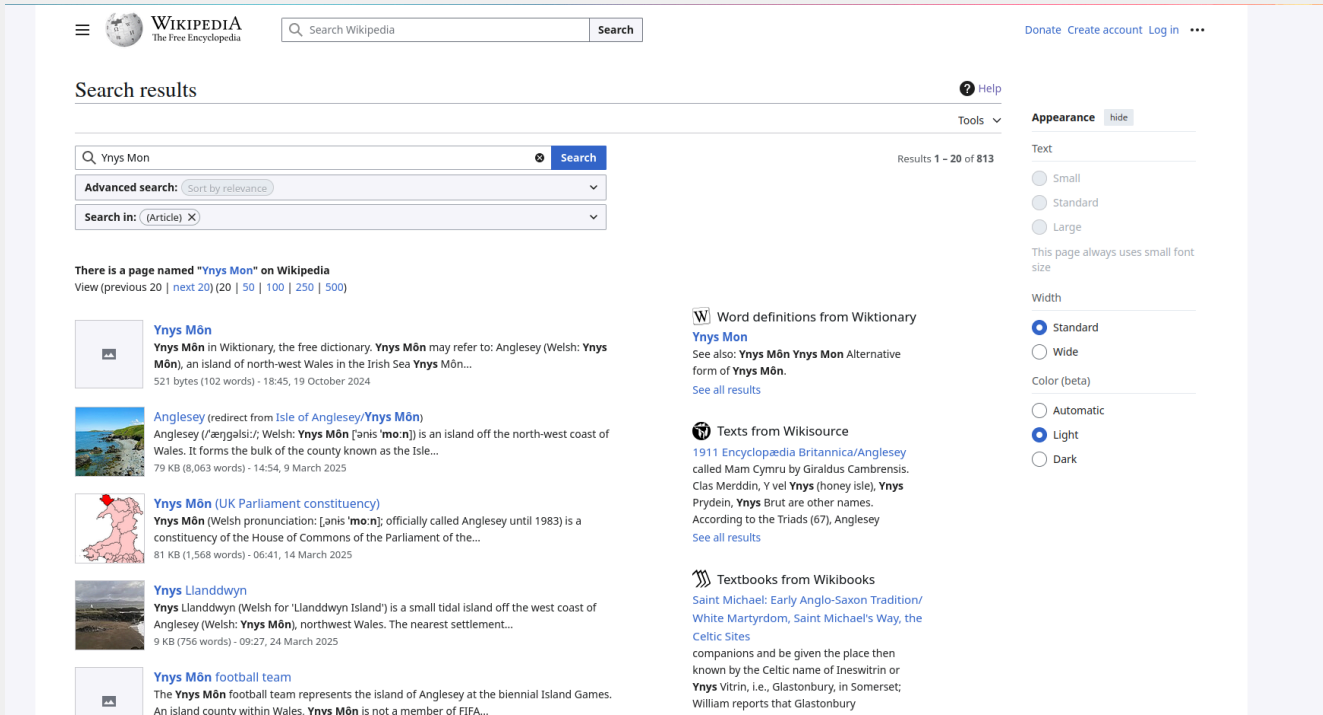
Key takeaways:

- Search box should always be visible
- Making the search field longer prompts users to make longer queries, which leads to better results.
- Spell Check and suggestions on the search bar help users search better.

EXISTING OUTCOMES



The New York Times uses a search engine to search its articles going back to 1831. It seems to only search the headlines, which is good enough for them since they have so many articles on so many things (13 million and counting), so any relevant query will probably have a headline associated with it. They have a clean UI, although I'm not a big fan about how the pictures are placed.



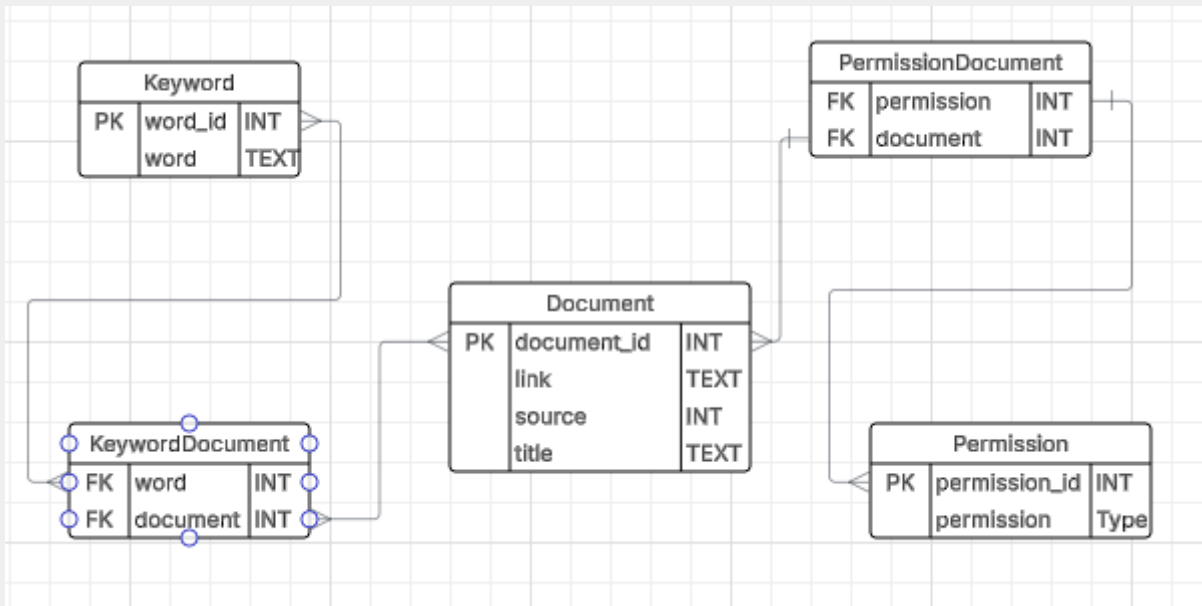
Wikipedia has its own search indexed engine, although it only uses it in the advanced search, since there's an article for nearly everything on wikipedia, they can just search the titles. The advanced search has a neat feature where the parts of your query that are relevant to the prompt are shown in a preview and bolded. The UI is very cluttered, but somehow still very ordered. It feels very professional.

Idea Generation

Colour palette taken from the schools website:



Basic layout for the database (Is very basic right now, might implement a cache for common searches):



Initial Search Page



Burnside High School  
Te Kura o Waimairi-iri

 Search Burnside...

Recent Searches

You have no recent searches!



Burnside High School

Te Kura o Waimairi-iri



How to call in sick

---

527 results found...



How to manage sick days

Staff support directory

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Absence Form

Staff support directory

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Report your child's absence

Parent Portal

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Reporting Absences

Parent Portal

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Layout for results page

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## Admin Panel

### Documents

### Users

### Permissions

### Logs

 Search document by Title..

2 results...

**Enrol | Burnside High School**

Application for Enrolment 2025

### Enrol | Burnside High School

Source: ww.burnside.school.nz

File type: Link

#### Permissions:

 Everyone





## Add Documents:

### Admin Panel FEEDBACK

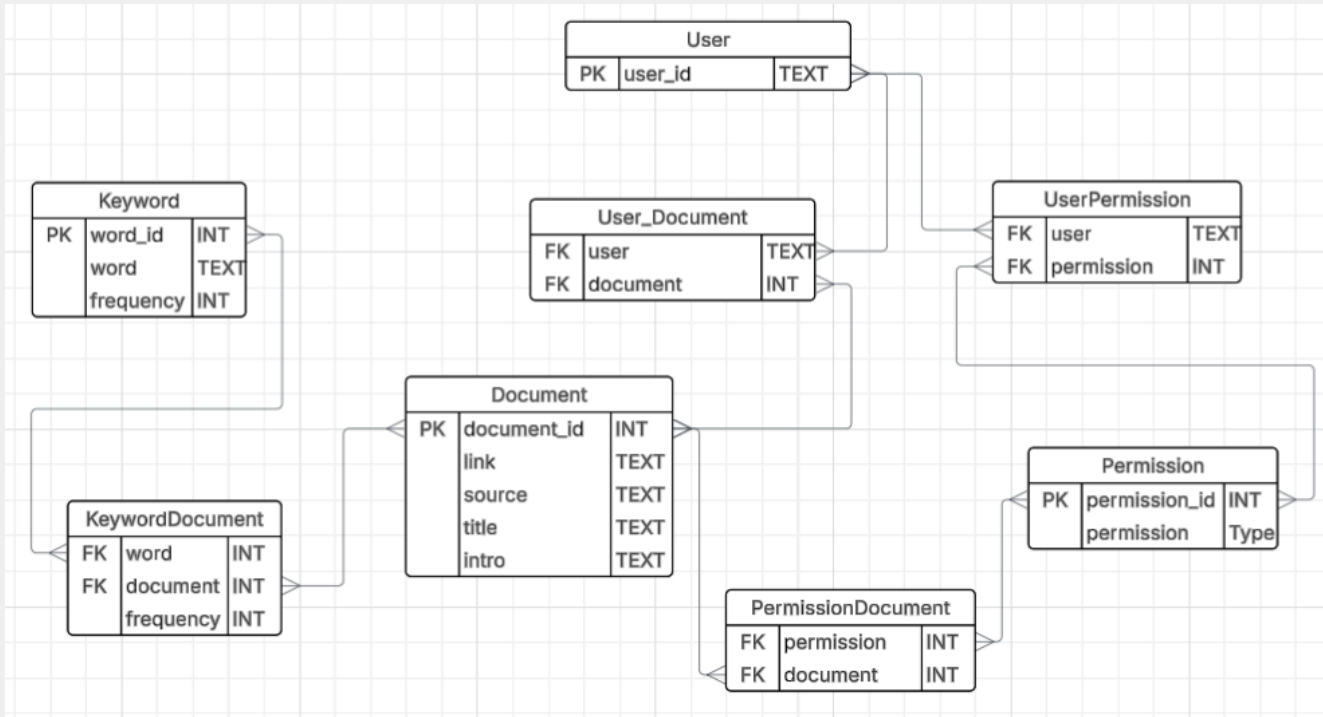
I was told that the colour palette was drab by my classmates. They suggested a colour palette similar to this:

|                 |                   |                           |                         |
|-----------------|-------------------|---------------------------|-------------------------|
| FFFFFF<br>White | 50C878<br>Emerald | 325044<br>Brunswick green | 173248<br>Prussian blue |
|-----------------|-------------------|---------------------------|-------------------------|

I got some more credible feedback, and they liked the first colour palette much more than this one. I have decided to just stick to the first one, since it's the colours on the school website

I talked to an experienced web developer (My Mum), and she for the most part liked the design. However she had a critique that the icons for the results were too ambiguous, if they're pdfs or web documents. She also suggested that the recent searches tab should be available on the searches page.

## Refinement



I added some tables for users and what documents they've previously searched so that it can be kept track of. I also added an intro column for each document so that the intro can be displayed in the search results.



Burnside High School  
Te Kura o Waimairi-iri

# How to call in sick

527 results found...



## How to manage sick days

Staff support directory

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## Absence Form

Staff support directory

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## Report your childs absence

Parent Portal

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## Reporting Absences

Parent Portal

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Recent Searches

I changed the icons to be more clear as to what the result is linking towards. I decided that the website icon should be a chain because the school uses a service called schoology for teachers

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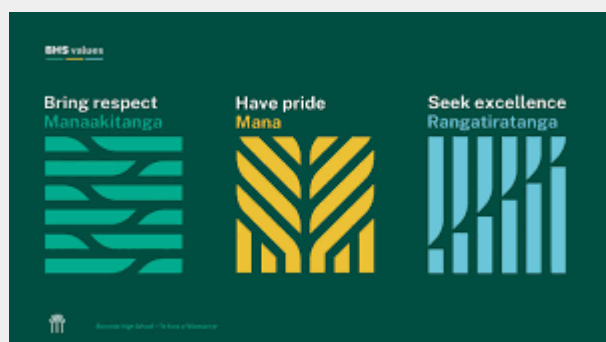
to provide resources to students, and that uses a chain to designate chains to mean website links.



Schoology Website Link

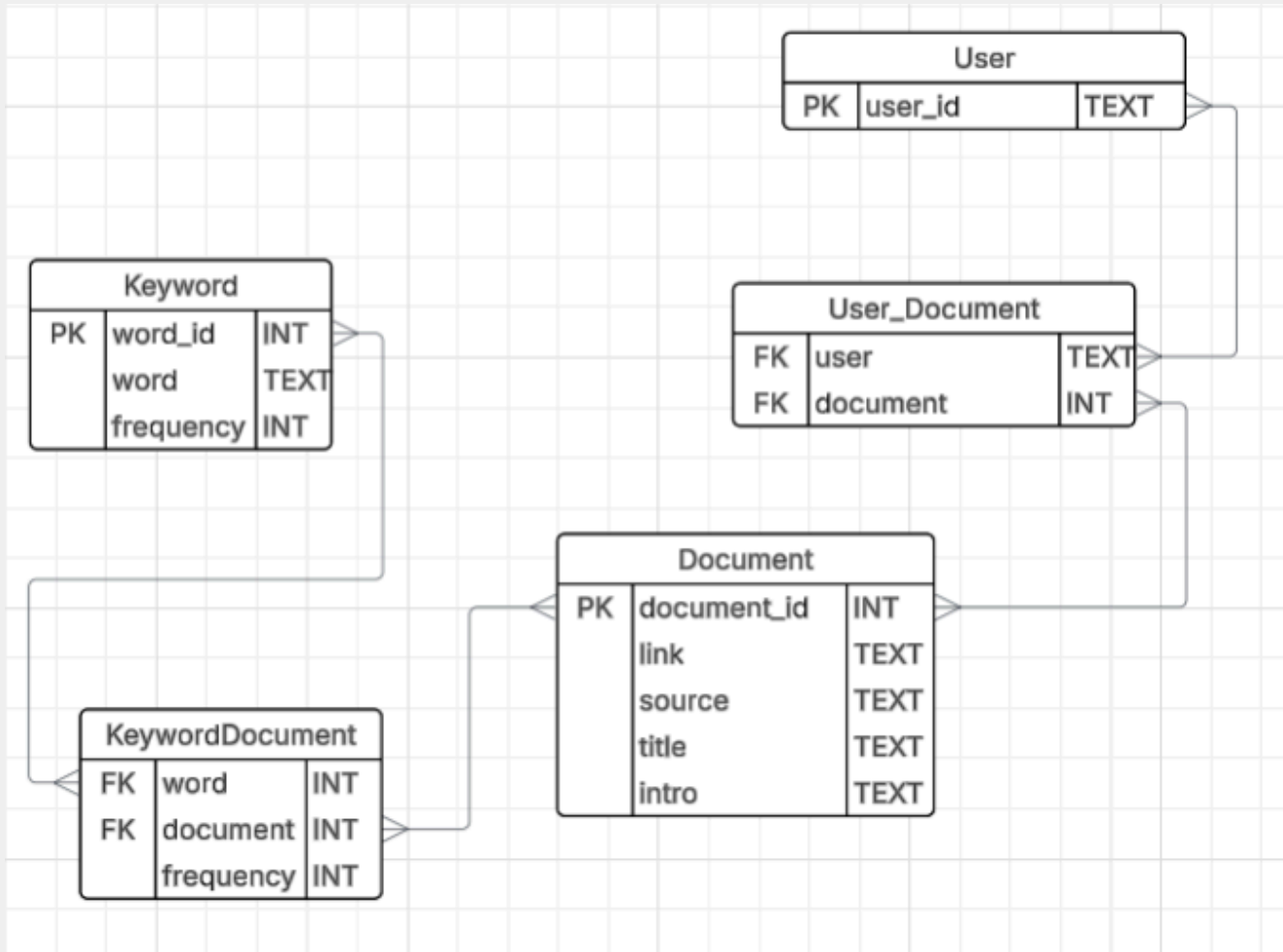
## FEEDBACK

I got feedback from 2 of the Digital Tech Teachers (Mr. Creighton and Mrs. Maddaford) at our school on the website and specifically how I plan to scrape documents. They were happy enough with the user design, although they suggested adding 3 extra designs that the school uses to represent the school values on the initial search landing page.



On the front of how I'm scraping and storing documents, they pointed out that it would be a lot of work to manually check for 100's of thousands of documents what the permissions are. They said that it would be better to scrap the idea of letting the website be open to students, to remove the complexity of having to store permissions and manage the permissions

## FINAL DESIGN



REmoved table containing permissions to reflect dropping permission functionality of website.

|        |        |                 |               |
|--------|--------|-----------------|---------------|
| FFFFFF | 64A188 | 325044          | 173248        |
| White  | Zomp   | Brunswick green | Prussian blue |





Burnside High School  
Te Kura o Waimairi-iri

Recent Searches

You have no recent searches!

 Search Burnside...



Burnside High School  
Te Kura o Waimairi-iri

How to call in sick

527 results found...



How to manage sick days

Staff support directory

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Absence Form

Staff support directory

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Report your childs absence

Parent Portal

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Reporting Absences

Parent Portal

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Recent Searches

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NIELSEN HEURISTICS

Two of the Nielsen Heuristics in this Design are:

- Aesthetic and Minimalist Design:
  - The design is very minimalist, drawing the user’s attention to the big search bar in the search page.
- Match between real world and System:
  - I use a magnifying glass to denote the search bar, which matches common symbols a user might see.

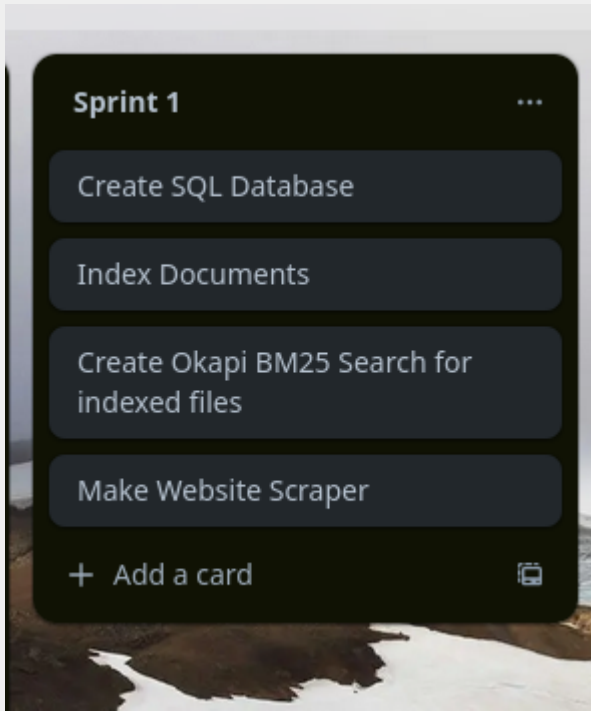
DEVELOPMENT

Development Stages Overview

| Key Stage | Aim & Description                                                                                                            | Due Date  |
|-----------|------------------------------------------------------------------------------------------------------------------------------|-----------|
| Sprint 1  | The aim of this sprint is to get a system in place to scan documents, websites, and html code, for the text content with in. | June      |
| Sprint 2  | The aim of this sprint is to have the search system functional so a user can search files that have been scraped             | August    |
| Sprint 3  | The aim of this sprint is to add styling and make the website more secure (the website will be handling sensitive            | September |

Sprint 1 - MVP (minimum viable product)  
Basic Framework / Functionality

I PLAN TO



WHAT I HAVE DEVELOPED DURING THIS SPRINT (the trialling)

|     |     |         |      |
|-----|-----|---------|------|
| 109 | 109 | these   | 682  |
| 110 | 110 | valuabl | 42   |
| 111 | 111 | some    | 547  |
| 112 | 112 | shine   | 11   |
| 113 | 113 | in      | 6430 |
| 114 | 114 | on      | 2567 |
| 115 | 115 | field   | 126  |
| 116 | 116 | emphasi | 65   |
| 117 | 117 | is      | 4913 |
| 118 | 118 | deliv   | 90   |
| 119 | 119 | broad   | 45   |
| 120 | 120 | balanc  | 36   |
| 121 | 121 | ensur   | 676  |
| 122 | 122 | discov  | 36   |
| 123 | 123 | follow  | 589  |
| 124 | 124 | their   | 1477 |
| 125 | 125 | passion | 58   |
| 126 | 126 | you     | 4434 |
| 127 | 127 | belong  | 119  |
| 128 | 128 | nurtur  | 27   |
| 129 | 129 | talent  | 75   |
| 130 | 130 | explor  | 181  |
| 131 | 131 | endless | 16   |
| 132 | 132 | possibl | 187  |
| 133 | 133 | await   | 12   |
| 134 | 134 | find    | 240  |

Made a system to scrape a web page for all the readable text (text not inside tag metadata or the such, and automatically upload each root word. An interesting thing to note is that I got more accurate word reductions from using snowball stemmer, the rigid stemmer that tries to use grammar rules to find root words, compared to lemmatization, which uses a big dictionary to find the root word for each possible word. This goes counter to my research and my own intuition that stemming isn't going to be very accurate because of how messy and full of edge cases the English language is. But for example, the word 'teaches' is kept as 'teaches' with lemmatization, but is turned to 'teach' with stemming. Going forward I will be using stemming instead of lemmatization.

## END OF SPRINT - WHAT I AM TESTING

Admin page to insert single documents and html files into database.

The screenshot shows a web application titled "Admin Panel". At the top, there is a navigation bar with four tabs: "Documents", "Users", "Permissions", and "Logs". The "Documents" tab is currently selected. Below the navigation bar, the main content area is titled "Insert Documents:". It contains a large, empty rectangular box with a thin border. Inside this box, there is a small, dark rectangular button with the text "Please fill out this field." Below this box is a horizontal line, and then a smaller, empty rectangular box with a thin border. Below this box is a small, dark rectangular button with the text "INSERT". Below the "INSERT" button is another horizontal line, and then a larger, empty rectangular box with a thin border. Inside this box, there is a small, dark rectangular button with the text "Browse..." and a small, light gray rectangular box with the text "No file selected." to its right.

Very barebones, and not very user friendly

## CLIENT / STAKEHOLDER / EXPERT TESTING & FEEDBACK

Mr Creighton, Head of Technology Department - A student project like this wouldn't be the final solution. What would be really great is a proof of concept website that shows that getting all the school's files in the same place is worth it.

Mrs Maddaford, Product design Teacher -

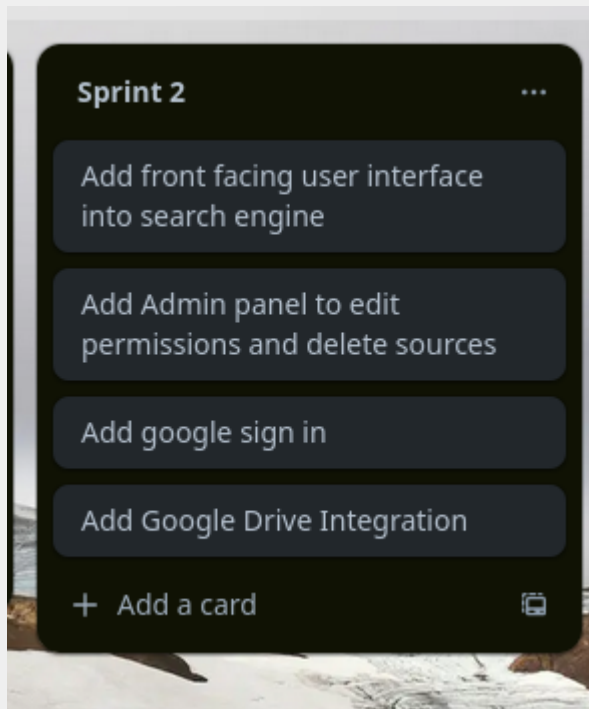
- Add a footer or border to cleanly finish the page design.
- Consider using school branding colours for visual consistency.

## NEXT STEPS

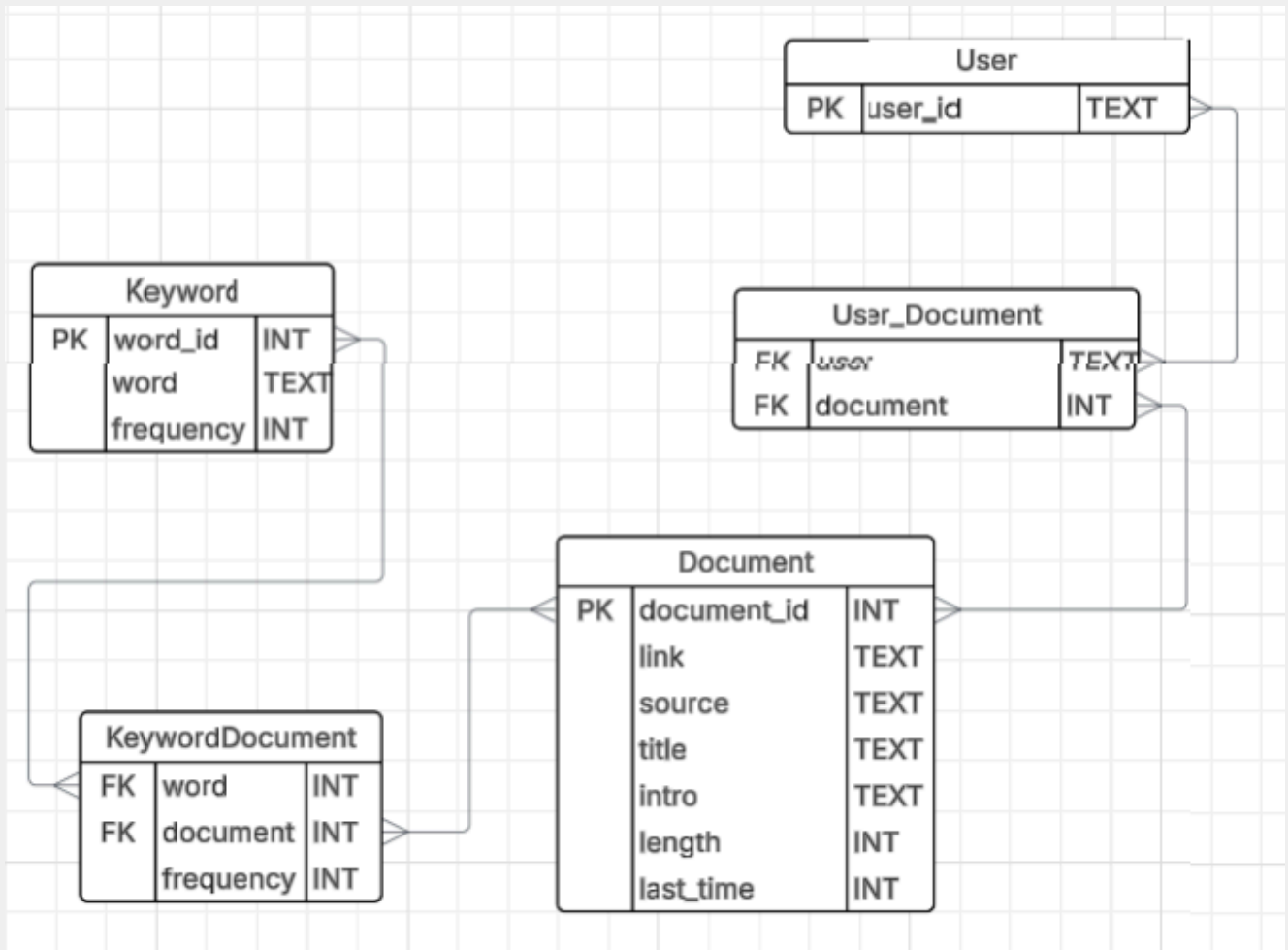
I now need to focus on broadening the scope of what I can search.

## Sprint 2 - MMP (minimum marketable product) Adding Content / Aesthetics

### I PLAN TO



### WHAT I HAVE DEVELOPED DURING THIS SPRINT (the trialling):



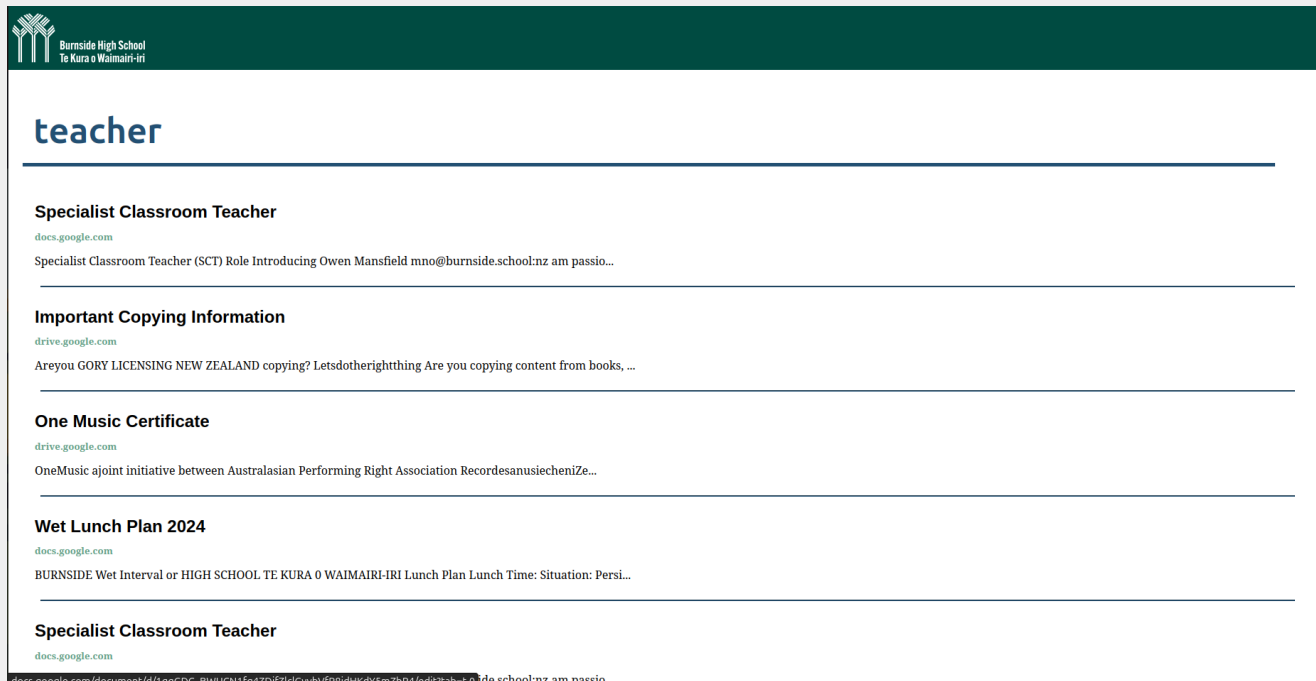
The document table was updated to include entries for the length of a document to help with the searching process by boosting documents that contain the same amount of keywords in a shorter length. Last\_time was also added to be a stopgap to prevent documents from being scraped again and again in quick succession, by first checking that the document hasn't already been scraped recently.

One of the important decisions I had to make was how I parse pdf's. I noticed that my initial pdf parser which was using a library called pypdf, was returning text with an inordinate amount of extra 0's. This turned out to be because pypdf was trying to guess words based on how close they were together in the raw data of the pdf (a pdf is really just a sequence of where different characters are on a page). If it didn't detect two letters being next to each other, it would return extra spaces. I looked into using OCR (optical character recognition), a kind of machine learning model that gets text from images. If I turned the pdf into an image, I could run OCR on it. I made a process to do it, but given my test machine was a laptop with no GPU, the machine learning model ran incredibly slow, often taking a minute just to read a single page. The machine learning should be sped up significantly when put on a server with an actual GPU, but for actual testing on my local machine, I have to use the more inaccurate pypdf. Luckily, the extra space issue seems to only be for pdf's exported from google drive.

Results Page

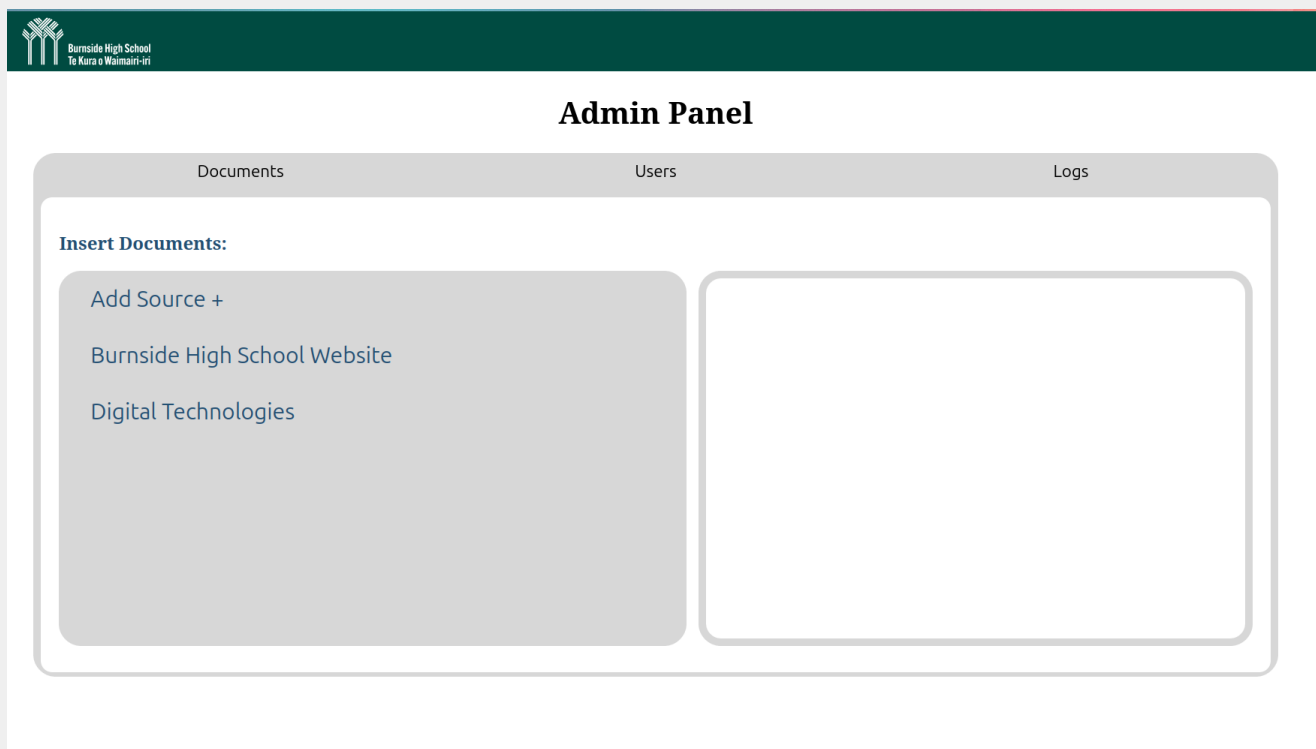
## END OF SPRINT - WHAT I AM TESTING

[images / or Video of what the outcome looks like at the end of this sprint]



The screenshot shows a web page with a dark green header containing the Burnside High School logo and name. Below the header, the word 'teacher' is displayed in a large, bold, blue font. The page lists several documents, each with a title, a Google Drive link, and a brief description. The documents are: 'Specialist Classroom Teacher' (SCT Role Introducing Owen Mansfield), 'Important Copying Information' (Are you copying content from books?), 'One Music Certificate' (OneMusic ajoint initiative), 'Wet Lunch Plan 2024' (Wet Interval or HIGH SCHOOL TE KURA 0 WAIMAIRI-IRI Lunch Plan), and another 'Specialist Classroom Teacher' document. The page is divided into sections by horizontal lines.

Results page that shows source, preview, and title



The screenshot shows the 'Admin Panel' interface. At the top, there is a dark green header with the Burnside High School logo and name. Below the header, the title 'Admin Panel' is centered. The main content area is divided into three tabs: 'Documents', 'Users', and 'Logs'. The 'Documents' tab is active, showing a section titled 'Insert Documents:'. Below this, there is a list of sources: 'Add Source +', 'Burnside High School Website', and 'Digital Technologies'. To the right of the list, there is a large, empty rectangular box, likely for a preview or additional information.

Updated admin page

### CLIENT / STAKEHOLDER / EXPERT TESTING & FEEDBACK

Mr Haines (School Principal): What you have looks great, but we want to be able to scrape files automatically from our google drive, so that an admin wouldn't have to do much.

Mr Sutton (School Network Administrator): I don't see the point in scraping google drive documents. Google Drive offers its own full text search for a user's files. Also, there's security concerns with giving a student's app access to confidential information like payment.

## NEXT STEPS

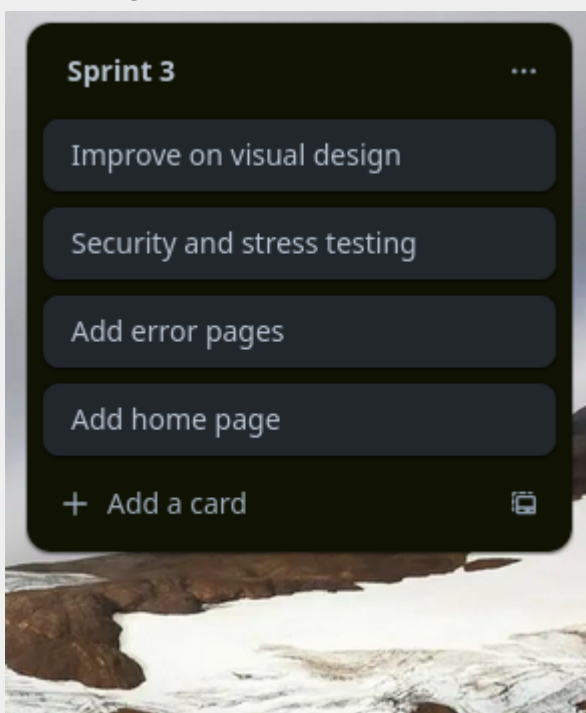
I had been putting off scraping the school's google drive because it looked like a huge hurdle. Interfacing with a third party's system (Google's Drive) and getting information out of it with . Mr Sutton's advice about Google drive is very useful, as I can offload some of my searching capability onto google drive.

---

## Sprint 3 - MLP / MDP / MAP (minimum loveable/delightful/awesome product)

### Extras & Polishing

#### I PLAN TO



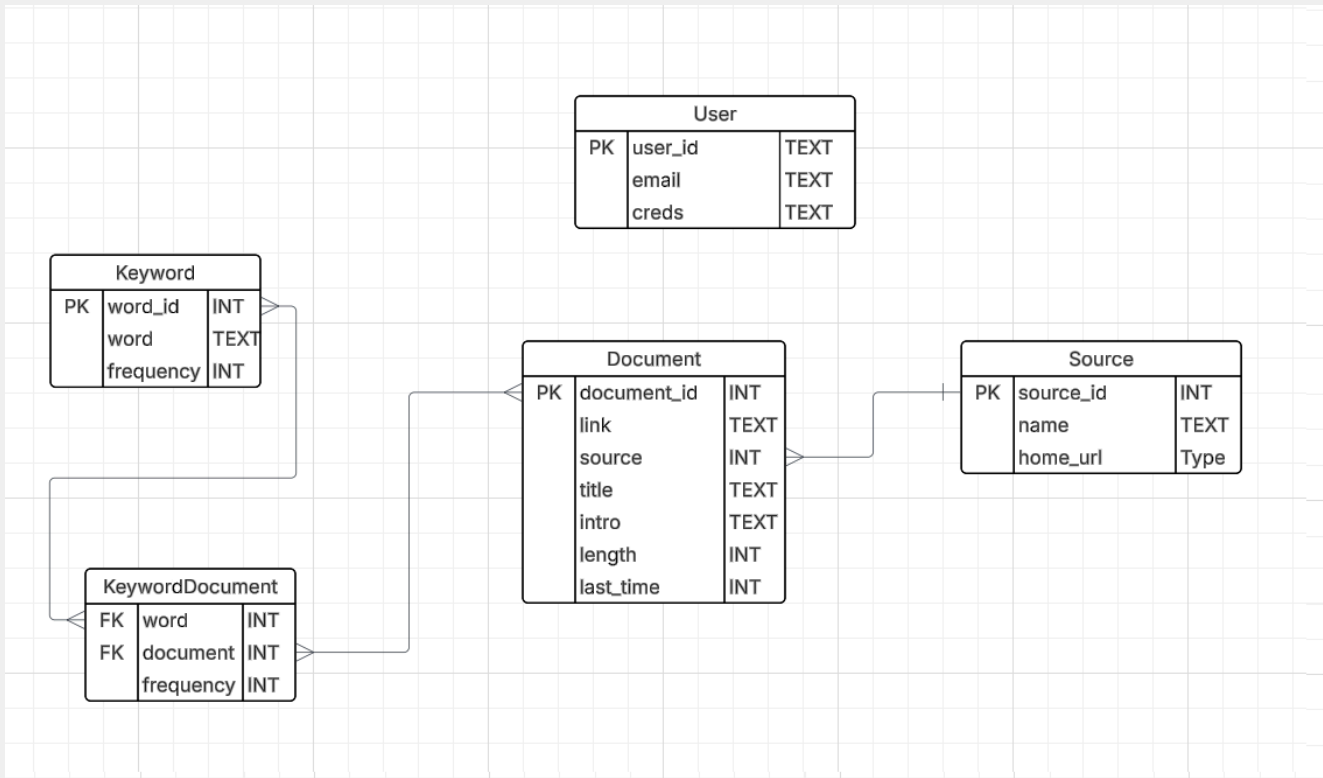
My main goal is make the website look much better, adding more of Burnside's symbols, and focusing on securing the website.

## WHAT I HAVE DEVELOPED IN THIS SPRINT (the trialling)

```
# Route to prevent Cross Site Scripting. Borrowed from
# https://stackoverflow.com/questions/63290047/flask-csp-content-security-policy-best-practice-against-attack-such-as-cross
@app.after_request
def add_security_headers(resp):
    resp.headers["Content-Security-Policy"] = (
        "default-src 'self'; "
        "script-src 'self' https://accounts.google.com/gsi/client https://code.jquery.com/jquery-3.6.0.min.js;"
        "frame-src https://accounts.google.com/gsi/;"
        "connect-src 'self' https://accounts.google.com/gsi/;"
        "img-src 'self' https://accounts.google.com/gsi/;"
        "style-src 'self' https://accounts.google.com/gsi/style"
    )
    return resp
```



Protection against XSS attacks by using Content Security Policy Headers



Removed a map table between User and Document as implementing search history is not mission critical, and I don't have enough time to implement it and also more important features.

END OF SPRINT - WHAT I AM TESTING



Burnside High School

To Kura o Waimairi-iri

Aditya Bansal

Creative Arts | Burnside High School

[Burnside High School Website](#)

Creative Arts | Burnside High School Skip to content Our School Our People Student Life Support Enro...

Term Dates | Burnside High School

[Burnside High School Website](#)

Term Dates | Burnside High School Skip to content Our School Our People Student Life Support Enrolme...

Specialist Music Programme | Burnside High School

[Burnside High School Website](#)

Specialist Music Programme | Burnside High School Skip to content Our School Our People Student Life...

Homepage | Burnside High School

[Burnside High School Website](#)

Homepage | Burnside High School Skip to content Our School Our People Student Life Support Enrolment...

Our Curriculum | Burnside High School

[Burnside High School Website](#)

Our Curriculum | Burnside High School Skip to content Our School Our People Student Life Support Enr...

Enrol | Burnside High School

[Burnside High School Website](#)

Enrol | Burnside High School Skip to content Our School Our People Student Life Support Enrolment In...



Changed around colour of source and added burnside band to the right to further push school branding, based on advice from product design teacher.

END USER TESTING & FEEDBACK

| Person                         | Feedback                                                                                                                                                                                                                                                                                                                        | What I'm going to do about it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ephram Chan (Non-Tech)         | The results page should have a filter on it by source                                                                                                                                                                                                                                                                           | I added a filter option to the search results page. This also increases accuracy of search results for the user depending on what the user wants.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| William Cochrane (Tech)        | I don't like how I have to sign in twice to access the website. It feels clunky                                                                                                                                                                                                                                                 | Google's official advice is that authorization and authentication processes should be separate, which is why there were two logins. However, I agree that it feels very clunky, so I went and changed the process so that the user only does the authorization process, and that counts as authentication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Mrs Maddaford                  | <p>I think the user experience would be better if there weren't two pages in the search process, the search page and the result page. Less friction if the user went straight to the result page.</p> <p>Another thing is that it might be useful if there's a recommended queries option for users who may have difficulty</p> | <p>On the note of recommended search queries, while it's a very user friendly and good idea, big companies like Google and the New York Times, or large projects like Wikipedia, are able to make recommended queries due to having a large volume of user searches to process for patterns in what a user wants, and or having all the data in proprietary databases where phrases can more easily be searched for, and thus recommended to the user.</p> <p>For the suggestion that the search page and result page be merged, I think it's a good idea, but I don't have the time to change around the structure for it. This project is continuing after submission, and I will be implementing this to streamline the searching process. This will also let users filter searches before searching.</p> |
| My father, (Systems Architect) | The functionality looks great,                                                                                                                                                                                                                                                                                                  | Both great suggestions,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

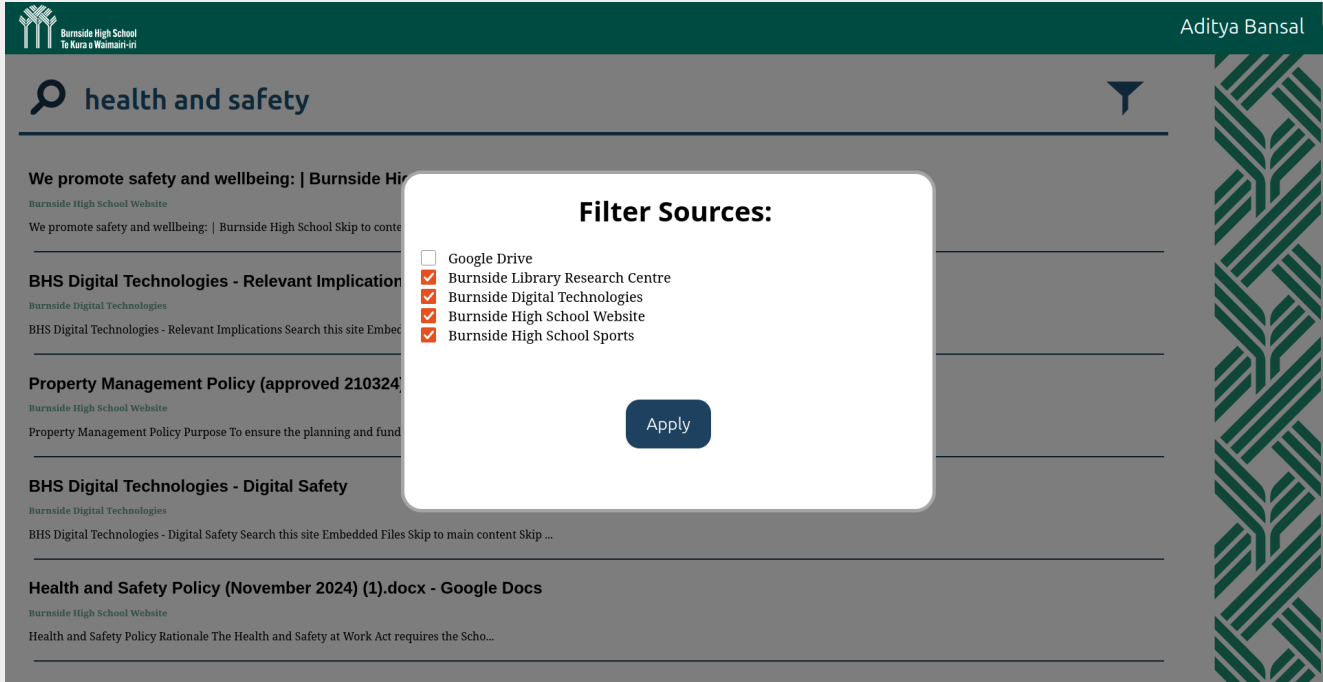
|  |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                 |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|  | <p>but the UX could use some work. The search page is very bare, you should move the search bar up and add something to the bottom like recent searches or how to use the website.</p> <p>The filter should have more options to filter by file type like pdf or page, and it should have an x button separate from the apply button</p> | <p>especially about making the filter more usable. I should take note of it for future development to make the feature that</p> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|

## FINAL DIGITAL TECHNOLOGIES OUTCOME

The screenshot shows a web application interface with a search page. The page has a search bar and a list of results. Below the page, a terminal window displays network requests and responses, including a successful login request and a search request. The terminal output shows the following requests and responses:

```
127.0.0.1 - - [18/Sep/2025 18:33:36] "GET /static/home.js HTTP/1.1" 304 -
127.0.0.1 - - [18/Sep/2025 18:33:36] "GET /static/BMS.svg HTTP/1.1" 304 -
127.0.0.1 - - [18/Sep/2025 18:33:36] "GET /static/backgrounds-cropped.svg HTTP/1.1" 304 -
Redirecting user to: https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=456948487872-fhaek6f6ev77q69fb54vqhiv8bgs5.apps.googleusercontent.com&redirect_uri=http%3A%2F%2Flocalhost%3A5000%2Fcallback&scope=https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fdrive.metadata.readonly=https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.profile=openId=https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email=state=msj5xso754p4R6pR3bQkV2LVcode=4&AAV6eR2DtpG8eyY43e0Jn0LxtyWQ0_FEIGH-APX5UMLr1C3aIV-3eJ8jccwv198FA6scope=email%3Dprofile%3Dhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fdrive.metadata.readonly%3Dhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.profile%3DopenId%3Dhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email%3Duser=26nd-burnsid-e.school.n2&prompt=consent HTTP/1.1" 302 -
127.0.0.1 - - [18/Sep/2025 18:59:34] "GET /search HTTP/1.1" 200 -
127.0.0.1 - - [18/Sep/2025 18:59:34] "GET /static/dropdown.js HTTP/1.1" 304 -
127.0.0.1 - - [18/Sep/2025 18:59:34] "GET /static/search.js HTTP/1.1" 304 -
127.0.0.1 - - [18/Sep/2025 18:59:34] "GET /static/BMS.svg HTTP/1.1" 304 -
127.0.0.1 - - [18/Sep/2025 18:59:34] "GET /static/background-cropped.svg HTTP/1.1" 304 -
```

Page 28 of 41

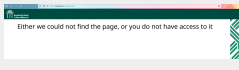
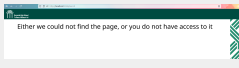
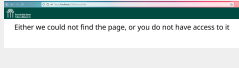
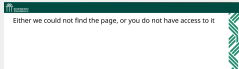
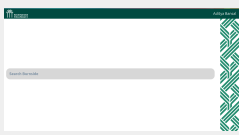
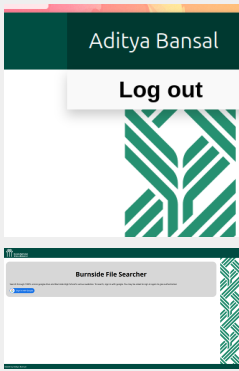
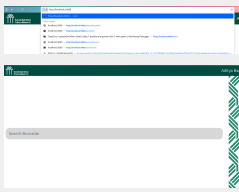
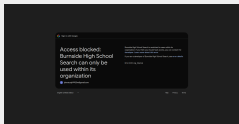


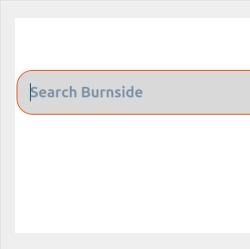
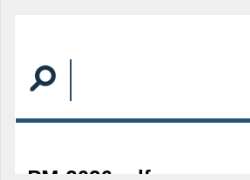
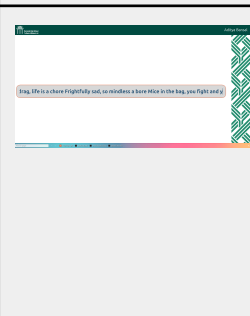
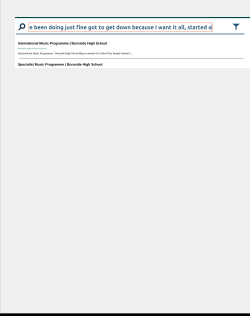
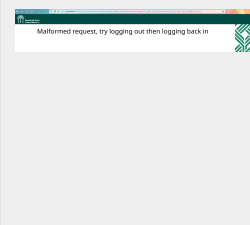
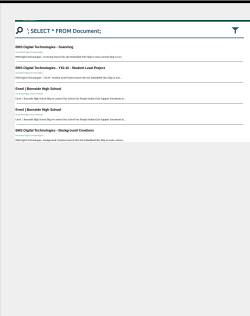
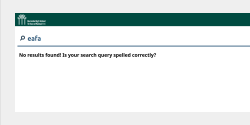
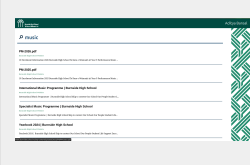
I added a filter for search queries which helps an experienced user filter for documents they want, which is an example of **Flexibility and Efficiency of use**

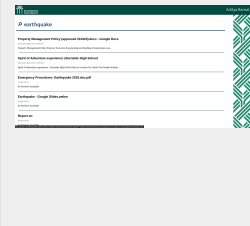
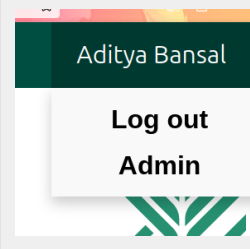
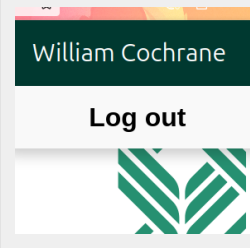
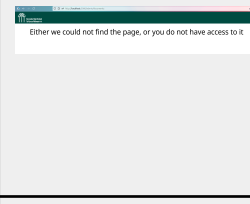
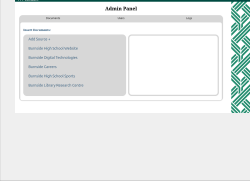
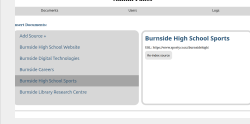
RELEVANT IMPLICATIONS

| Implication                              | Evidence of addressal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| End User Considerations (Website)        | The site uses google logins to authenticate users. The intended end users are teachers at Burnside High School, who have google accounts. This should be more convenient for them.<br>(end user considerations for website)                                                                                                                                                                                                                                                                                          |
| Security (Database)                      | <ul style="list-style-type: none"><li>• I've added protections for cross site scripting attacks to stop attackers from running their own script on a website.</li><li>• I've set up the scopes for my google drive API access so that even if an attacker gets their way onto the website, the most that they can get out of a user is the metadata of their files.</li><li>• The most sensitive data (API access tokens) are stored at rest and aren't transmitted over the web</li></ul><br>(Security for website) |
| Futureproofing/Sustainability (Database) | My website has an admin page so that sources can be added in through a GUI without having to interface with the code. (future proofing data entry for the database)                                                                                                                                                                                                                                                                                                                                                  |
| Usability (Website)                      | I addressed the problem of having to sign in twice, which makes the website                                                                                                                                                                                                                                                                                                                                                                                                                                          |

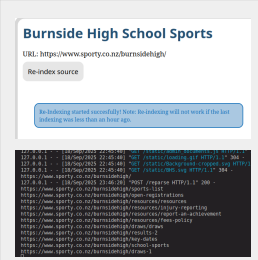
TESTING

| Test                                    | Expected Behaviour                                     | Observed Behaviour | Evidence                                                                             | Notes                                                                               |
|-----------------------------------------|--------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 404 Handling                            | User redirected to 404 Page                            | Expected           |    |                                                                                     |
| 403 Handling - Search Page              | User directed to 403 Page                              | Expected           |    | 403 and 404 page look identical on purpose to deter attackers                       |
| 403 Handling - Results Page             | User directed to 403 Page                              | Expected           |    |                                                                                     |
| 403 Handling - Admin Page               | User Directed to 403 page                              | Expected           |    |                                                                                     |
| Google Login with Burnside              | Take user to search page signed in                     | Expected           |   | <a href="mailto:23533@burnside.school.nz">23533@burnside.school.nz</a> login used   |
| Log out button                          | Take user back to search page logged out               | Expected           |  |                                                                                     |
| Home page redirect                      | Home page redirects user to search when user logged in | Expected           |  |                                                                                     |
| Prevent non school users from accessing | Google authentication blocks user                      | Expected           |  | <a href="mailto:johnstudy19925w@gmail.com">johnstudy19925w@gmail.com</a> login used |
| Search Functionality                    |                                                        |                    |                                                                                      |                                                                                     |

|                                       |                                                                                                                     |          |                                                                                      |                                                                                                        |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Blank query prevention - Search Page  | Search page does not redirect                                                                                       | Expected |    | I am pressing enter in the example                                                                     |
| Blank query prevention - Results Page | Results page does not redirect                                                                                      | Expected |    | Pressing enter in the example                                                                          |
| Long query prevention - Search Page   | Bar does not allow inputting more than 100 characters                                                               | Expected |    | Copy pasted very long text into the search bar, and tried pressing more characters, capped out at 100. |
| Long Query Prevention - Results Page  | Bar does not allow inputting more than 100 characters                                                               | Expected |   | Copy pasted very long text into the search bar, and tried pressing more characters, capped out at 100. |
| Long Query Prevention - URL           | Trying to add more characters into query returns bad request code                                                   | Expected |  | Did a normal search query at max, then added more characters to see what would happen                  |
| Search Bar SQL Injection Prevention   | No effect, all non alphanumeric characters are stripped from query before being passed into searching SQL Database. | Expected |  |                                                                                                        |
| No results page                       | Query with no results returns no results page                                                                       | Expected |  |                                                                                                        |
| Generic Query: 1                      | Page returns with relevant results when broad search                                                                | Expected |  | Test 1                                                                                                 |

|                                                      |                                                                                                      |          |                                                                                      |        |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------|--------|
|                                                      | term: "Music"                                                                                        |          |                                                                                      |        |
| Generic Query: 2                                     | Page returns with relevant results when broad search term: "Earthquake"                              | Expected |    | Test 2 |
| Generic Query: 3                                     | Page returns relevant results when broad search term: "Health and Safety"                            | Expected |    | Test 3 |
| Blank term in url                                    | User is redirected to search page                                                                    | Expected |    |        |
| Admin Page                                           |                                                                                                      |          |                                                                                      |        |
| Admin page link showing up for admin account         | Admin page link shows up for admin user 23533@burnside.school.nz                                     | Expected |   |        |
| Admin page link not showing up for non admin account | Admin page link shows up for non admin user 21339@burnside.school.nz                                 | Expected |  |        |
| Admin page blocked from access by non admin account. | 403 error returned for non admin user 21339@burnside.school.nz                                       | Expected |  |        |
| Admin Page opens for admin account                   | Admin Page returned for admin <a href="mailto:23533@burnside.school.nz">23533@burnside.school.nz</a> | Expected |  |        |
| Display Source                                       | Info for source is displayed when clicking button                                                    | Expected |  |        |



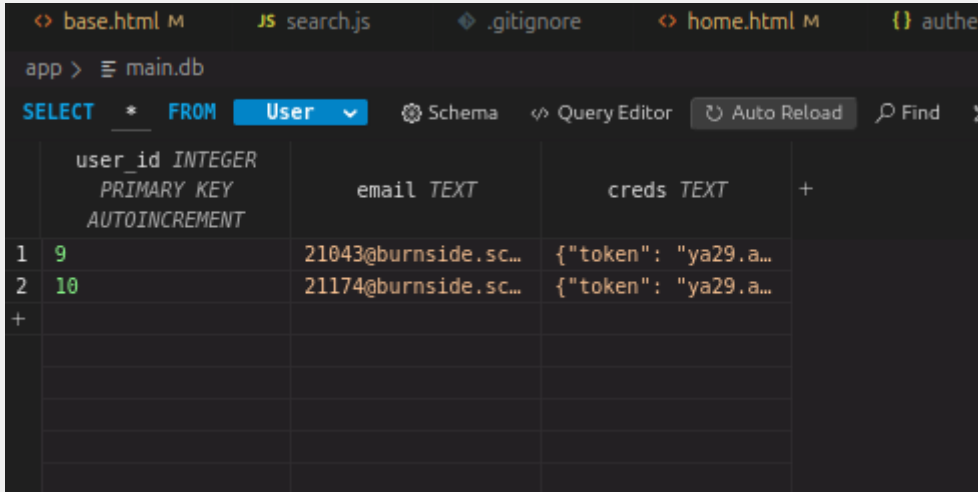
|                 |                                                                                                           |          |                                                                                    |  |
|-----------------|-----------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------|--|
| Reindex trigger | When reindex button is pressed, console reflects that website is being reindexed after /reparse is called | Expected |  |  |
|-----------------|-----------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------|--|

Tests for adding new source in Data integrity

DATA INTEGRITY

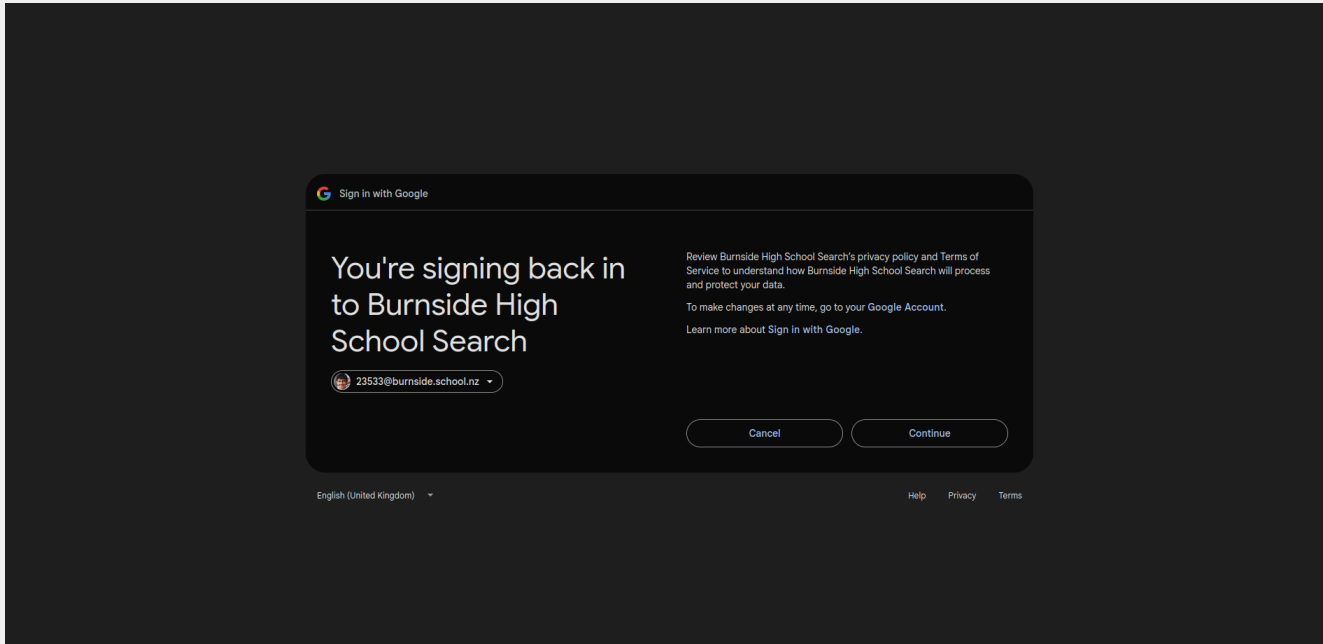
User Table:

User table before login:



|   | user_id INTEGER<br>PRIMARY KEY<br>AUTOINCREMENT | email TEXT           | creds TEXT           | + |
|---|-------------------------------------------------|----------------------|----------------------|---|
| 1 | 9                                               | 21043@burnside.sc... | {"token": "ya29.a... |   |
| 2 | 10                                              | 21174@burnside.sc... | {"token": "ya29.a... |   |
| + |                                                 |                      |                      |   |
|   |                                                 |                      |                      |   |
|   |                                                 |                      |                      |   |
|   |                                                 |                      |                      |   |
|   |                                                 |                      |                      |   |
|   |                                                 |                      |                      |   |

Logging in with 23533@burnside.school.nz:



New user record added

SELECT \* FROM User Schema Query Editor Auto Reload

|   | user_id INTEGER<br>PRIMARY KEY<br>AUTOINCREMENT | email TEXT           | creds TEXT           | + |
|---|-------------------------------------------------|----------------------|----------------------|---|
| 1 | 9                                               | 21043@burnside.sc... | {"token": "ya29.a... |   |
| 2 | 10                                              | 21174@burnside.sc... | {"token": "ya29.a... |   |
| 3 | 11                                              | 23533@burnside.sc... | {"token": "ya29.a... |   |
| + |                                                 |                      |                      |   |
|   |                                                 |                      |                      |   |
|   |                                                 |                      |                      |   |
|   |                                                 |                      |                      |   |

Querying

There should be no change in the database. Querying function interacts with Document, KeywordDocument, and Keyword.

Tables before query

Keyword Table:

app > main.db

SELECT \* FROM Keyword Schema Query Editor Auto Reload

|      | word_id INTEGER<br>PRIMARY KEY | word TEXT   | frequency INT | + |
|------|--------------------------------|-------------|---------------|---|
| 9676 | 9675                           | unfinanci   | 1             |   |
| 9677 | 9676                           | creditdebit | 1             |   |
| 9678 | 9677                           | 3191        | 1             |   |
| 9679 | 9678                           | 0002255     | 1             |   |
| 9680 | 9680                           | zonta       | 1             |   |
| 9681 | 9681                           | interschool | 1             |   |
| 9682 | 9682                           | 3km         | 1             |   |
| 9683 | 9683                           | 45km        | 1             |   |
| 9684 | 9684                           | mayjun      | 1             |   |
| 9685 | 9685                           | talli       | 1             |   |
| 9686 | 9686                           | rahi        | 1             |   |
| 9687 | 9687                           | cando       | 1             |   |
| +    |                                |             |               |   |
|      |                                |             |               |   |
|      |                                |             |               |   |
|      |                                |             |               |   |
|      |                                |             |               |   |

## Document Table:

|     | document_id INTEGER<br>PRIMARY KEY | link TEXT                                         | source INTEGER<br>REFERENCES<br>Source(source_id) | title TEXT           | intro TEXT           | length INTEGER | last_time INTEGER  | + |
|-----|------------------------------------|---------------------------------------------------|---------------------------------------------------|----------------------|----------------------|----------------|--------------------|---|
| 473 | 474                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 477            | 1757398017.1663263 |   |
| 474 | 475                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 479            | 1757398030.5726867 |   |
| 475 | 476                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 301            | 1757398039.6527545 |   |
| 476 | 477                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 212            | 1757398046.7371433 |   |
| 477 | 478                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 312            | 1757398055.8097773 |   |
| 478 | 479                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 278            | 1757398064.4329987 |   |
| 479 | 480                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 555            | 1757398079.8229568 |   |
| 480 | 481                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 451            | 1757398092.198562  |   |
| 481 | 482                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 210            | 1757398098.6547103 |   |
| 482 | 483                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 570            | 1757398113.5336003 |   |
| 483 | 484                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 195            | 1757398120.6378386 |   |
| 484 | 485                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 547            | 1757399152.2320213 |   |
| 485 | 486                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 472            | 1757399165.7942202 |   |
| 486 | 487                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 171            | 1757399196.2013502 |   |
| 487 | 488                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 171            | 1757399202.1049294 |   |
| 488 | 489                                | <a href="https://www.bu...">https://www.bu...</a> | 1 name: 'Burnside_                                | Teacher of Health... | Teacher of Health... | 229            | 1757399127.9907815 |   |
| 489 | 490                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 221            | 1757886710.6029882 |   |
| 490 | 491                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 221            | 1757886720.2112083 |   |
| 491 | 492                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 898            | 1757886728.3584542 |   |
| 492 | 493                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 112            | 1757886752.3170035 |   |
| 493 | 494                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886756.5896478 |   |
| 494 | 495                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886759.78252   |   |
| 495 | 496                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886761.5651138 |   |
| 496 | 497                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 232            | 1757886766.1775026 |   |
| 497 | 498                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886772.959678  |   |
| 498 | 499                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886774.8593974 |   |
| 499 | 500                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 113            | 1757886777.0384386 |   |
| 500 | 501                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 219            | 1757886780.9161744 |   |
| +   |                                    |                                                   |                                                   |                      |                      |                |                    |   |

KeywordDocument Table:

[illegible]

Query



Burnside High School

To Kura o Waimairi-iti

Aditya Bansal

 health and safety

We promote safety and wellbeing: | Burnside High School

[Burnside High School Website](#)

We promote safety and wellbeing: | Burnside High School Skip to content Our School Our People Studen...

BHS Digital Technologies - Relevant Implications

[Burnside Digital Technologies](#)

BHS Digital Technologies - Relevant Implications Search this site Embedded Files Skip to main conten...

BHS Digital Technologies - Digital Safety

[Burnside Digital Technologies](#)

BHS Digital Technologies - Digital Safety Search this site Embedded Files Skip to main content Skip ...

Property Management Policy (approved 210324).docx - Google Docs

[Burnside High School Website](#)

Property Management Policy Purpose To ensure the planning and funding of maintenance an...

Health and Safety Policy (November 2024) (1).docx - Google Docs

[Burnside High School Website](#)

Health and Safety Policy Rationale The Health and Safety at Work Act requires the Scho...



Tables after Query

app > main.db

SELECT \* FROM Keyword Schema Query Editor Auto Reload

|      | word_id INTEGER<br>PRIMARY KEY | word TEXT   | frequency INT |  |
|------|--------------------------------|-------------|---------------|--|
| 9676 | 9675                           | unfinanci   | 1             |  |
| 9677 | 9676                           | creditdebit | 1             |  |
| 9678 | 9677                           | 3191        | 1             |  |
| 9679 | 9678                           | 0002255     | 1             |  |
| 9680 | 9680                           | zonta       | 1             |  |
| 9681 | 9681                           | interschool | 1             |  |
| 9682 | 9682                           | 3km         | 1             |  |
| 9683 | 9683                           | 45km        | 1             |  |
| 9684 | 9684                           | mayjun      | 1             |  |
| 9685 | 9685                           | talli       | 1             |  |
| 9686 | 9686                           | rahi        | 1             |  |
| 9687 | 9687                           | cando       | 1             |  |
| +    |                                |             |               |  |
|      |                                |             |               |  |
|      |                                |             |               |  |
|      |                                |             |               |  |
|      |                                |             |               |  |

|     | document_id INTEGER<br>PRIMARY KEY | link TEXT                                         | source INTEGER<br>REFERENCES<br>Source(source_id) | title TEXT           | intro TEXT           | length INTEGER | last_time INTEGER  | + |
|-----|------------------------------------|---------------------------------------------------|---------------------------------------------------|----------------------|----------------------|----------------|--------------------|---|
| 473 | 474                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 477            | 1757398017.1663263 |   |
| 474 | 475                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 479            | 1757398030.5726867 |   |
| 475 | 476                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 301            | 1757398039.6527545 |   |
| 476 | 477                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 212            | 1757398046.7371433 |   |
| 477 | 478                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 312            | 1757398055.8097773 |   |
| 478 | 479                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 278            | 1757398064.4329987 |   |
| 479 | 480                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 555            | 1757398079.8229568 |   |
| 480 | 481                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 451            | 1757398092.198562  |   |
| 481 | 482                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 210            | 1757398098.6547103 |   |
| 482 | 483                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 570            | 1757398113.5336003 |   |
| 483 | 484                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 195            | 1757398120.6378386 |   |
| 484 | 485                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 547            | 1757399152.2320213 |   |
| 485 | 486                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 472            | 1757399165.7942202 |   |
| 486 | 487                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 171            | 1757399196.2013502 |   |
| 487 | 488                                | <a href="https://sites...">https://sites...</a>   | 2 name: 'Burnside_                                | BHS Digital Techn... | BHS Digital Techn... | 171            | 1757399202.1049294 |   |
| 488 | 489                                | <a href="https://www.bu...">https://www.bu...</a> | 1 name: 'Burnside_                                | Teacher of Health... | Teacher of Health... | 229            | 1757399127.9907815 |   |
| 489 | 490                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 221            | 1757886710.6029882 |   |
| 490 | 491                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 221            | 1757886720.2112083 |   |
| 491 | 492                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 898            | 1757886728.3584542 |   |
| 492 | 493                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 112            | 1757886752.3170035 |   |
| 493 | 494                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886756.5896478 |   |
| 494 | 495                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886759.78252   |   |
| 495 | 496                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886761.5651138 |   |
| 496 | 497                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 232            | 1757886766.1775026 |   |
| 497 | 498                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886772.959678  |   |
| 498 | 499                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 55             | 1757886774.8593974 |   |
| 499 | 500                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 113            | 1757886777.0384386 |   |
| 500 | 501                                | <a href="https://www.sp...">https://www.sp...</a> | 4 name: 'Burnside_                                | Burnside High Sch... | Burnside High Sch... | 219            | 1757886780.9611744 |   |
| +   |                                    |                                                   |                                                   |                      |                      |                |                    |   |

[illegible]

## Source Table

Source table in db:

| source_id INTEGER<br>PRIMARY KEY | name TEXT            | home_url TEXT                                     | + |
|----------------------------------|----------------------|---------------------------------------------------|---|
| 1                                | Burnside High Sch... | <a href="https://www.bu...">https://www.bu...</a> |   |
| 2                                | Burnside Digital ... | <a href="https://sites...">https://sites...</a>   |   |
| 3                                | Burnside Careers     | <a href="https://burnsi...">https://burnsi...</a> |   |
| 4                                | Burnside High Sch... | <a href="https://www.sp...">https://www.sp...</a> |   |
|                                  |                      |                                                   |   |
|                                  |                      |                                                   |   |

Source table is displayed on the school website

Admin Panel

Documents

Users

Insert Documents:

Add Source +

Burnside High School Website

Burnside Digital Technologies

Burnside Careers

Burnside High School Sports

Add new source

Adding a new source should create a new table, and add a couple of documents, keywords, and keyword document maps:

Before new source:

Source

| source_id INTEGER PRIMARY KEY | name TEXT            | home_url TEXT                                         |
|-------------------------------|----------------------|-------------------------------------------------------|
| 1                             | Burnside High Sch... | <a href="https://www.bu...">https://www.bu...</a>     |
| 2                             | Burnside Digital ..  | <a href="https://sites.bu...">https://sites.bu...</a> |
| 3                             | Burnside Careers     | <a href="https://burnsi...">https://burnsi...</a>     |
| 4                             | Burnside High Sch... | <a href="https://www.sp...">https://www.sp...</a>     |
| 5                             | Burnside Library ..  | <a href="https://bhslib-...">https://bhslib-</a>      |

Document

| document_id INTEGER PRIMARY KEY | link TEXT                                        | source INTEGER REFERENCES Source(source_id) |
|---------------------------------|--------------------------------------------------|---------------------------------------------|
| 507                             | <a href="https://bhslib-...">https://bhslib-</a> | 5 name: 'Burnside...                        |
| 508                             | <a href="https://bhslib-...">https://bhslib-</a> | 5 name: 'Burnside...                        |
| 509                             | <a href="https://bhslib-...">https://bhslib-</a> | 5 name: 'Burnside...                        |
| 510                             | <a href="https://bhslib-...">https://bhslib-</a> | 5 name: 'Burnside...                        |
| 511                             | <a href="https://bhslib-...">https://bhslib-</a> | 5 name: 'Burnside...                        |

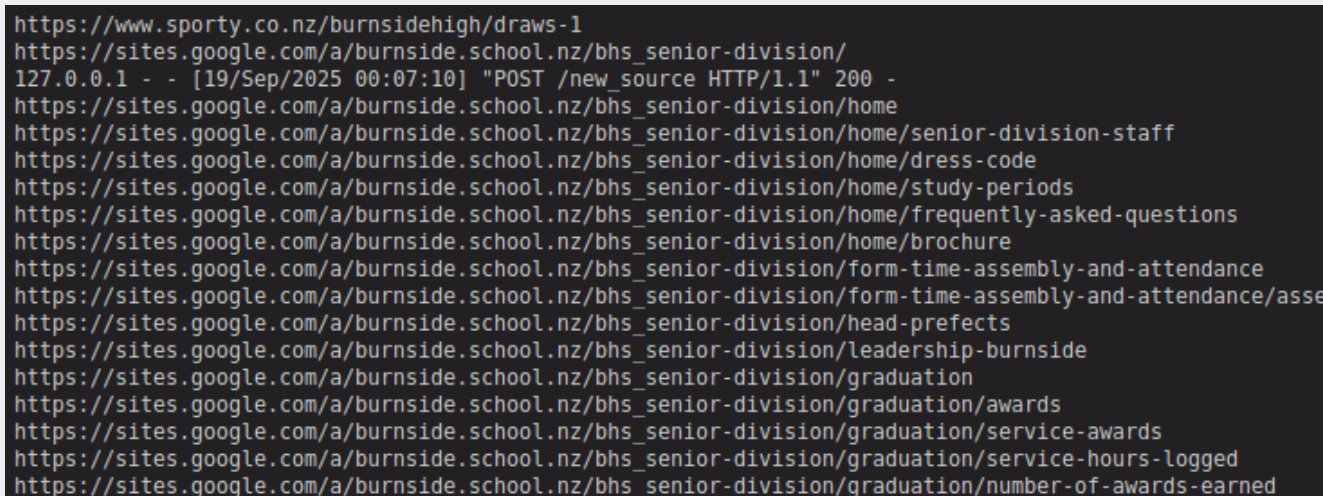
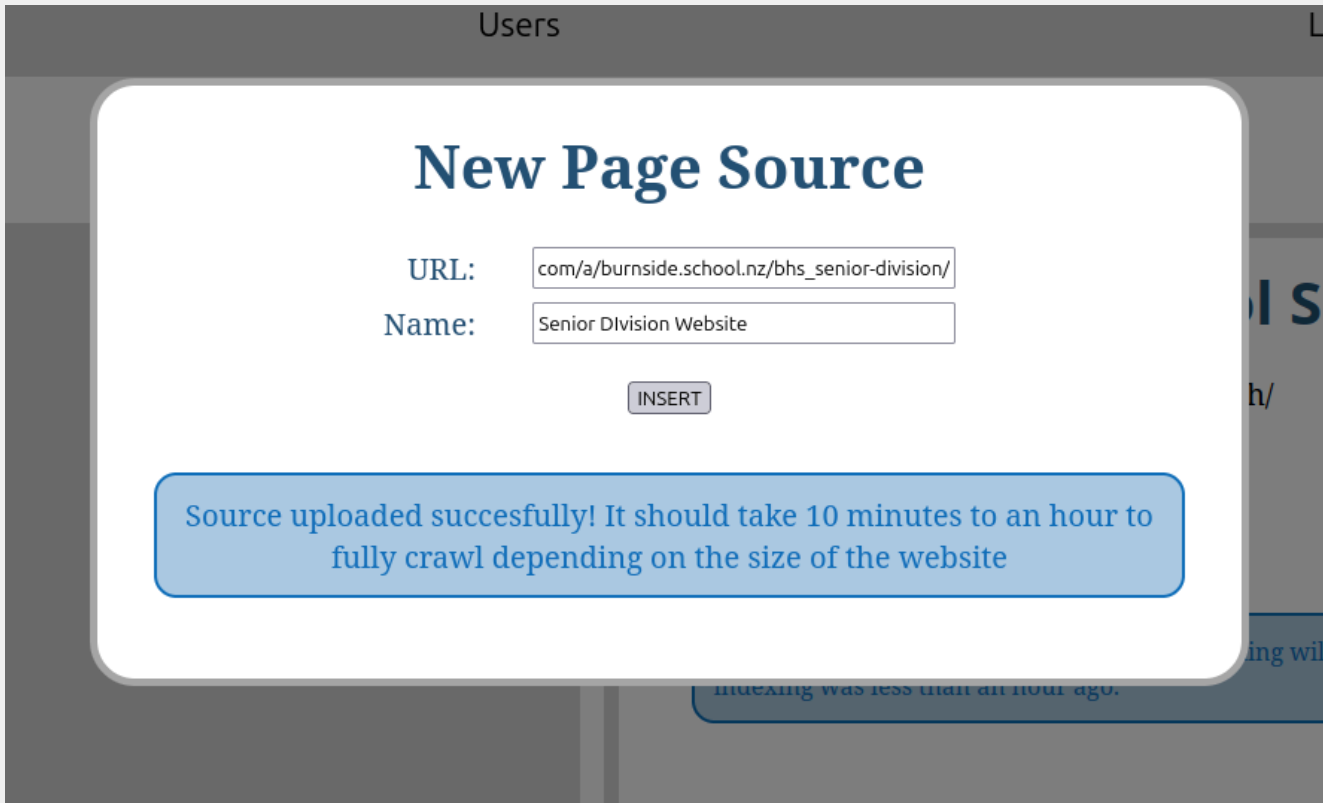
Keyword

| word_id INTEGER PRIMARY KEY | word TEXT | frequency INT |
|-----------------------------|-----------|---------------|
| 9825                        | canti     | 1             |
| 9826                        | uni       | 1             |
| 9827                        | bibm      | 1             |
| 9828                        | endnot    | 1             |

KeywordDocument

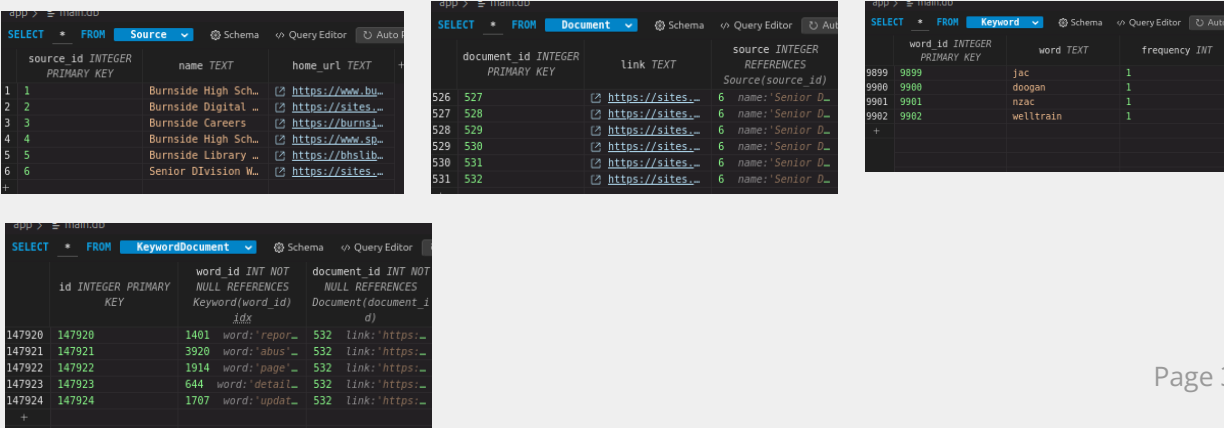
| id INTEGER PRIMARY KEY | word_id INT NOT NULL REFERENCES Keyword(word_id) | document_id INT NOT NULL REFERENCES Document(document_id) | frequency INT |
|------------------------|--------------------------------------------------|-----------------------------------------------------------|---------------|
| 144562                 | 199 word: 'with' ..                              | 511 Link: 'https:...                                      | 1             |
| 144563                 | 4782 word: 'custo...                             | 511 Link: 'https:...                                      | 1             |
| 144564                 | 2632 word: 'templ...                             | 511 Link: 'https:...                                      | 1             |
| 144565                 | 586 word: 'start' ..                             | 511 Link: 'https:...                                      | 1             |

Adding a new source:



The console shows all the pages being scraped

After new source:



There is new information in the Source, Document, Keyword, and KeywordDocument Table, so most likely pages have been indexed successfully.

Index Integrity:

Using the page <https://www.burnside.school.nz/absences/>

# Absences

---

**If your child is going to be absent for any reason it is essential that you let us know as soon as possible.**

It is important that you phone/text or email us to confirm your child's absence and provide us with the reason.

We acknowledge that there are genuine reasons students may occasionally be absent. However, without regular attendance your child will struggle to make progress. If your child is absent regularly, or for extended periods, we will reach out to you to discuss the situation.

We are committed to supporting your child to attend school. If you have difficulty getting your child to school, you can ask for support by emailing your child's form teacher or dean. We can work with you to develop a support plan that addresses any barriers to regular attendance. Regular attendance is considered having less than five days absent per term.

## Absentee procedures for Burnside High School

If your child is going to be absent or late please notify the school office in one of the following ways:

**Text:** [022 043 6951](tel:0220436951)  
**Phone:** [033588383](tel:033588383), extension 1  
**Email:** [absences@burnside.school.nz](mailto:absences@burnside.school.nz)  
**Notes:** Bring a note from home and give to the student office

When reporting an absence, kindly provide the following information:

**Your name**  
**Child's name and student number**  
**A complete reason for their absence (e.g dentist appointment)**

For more information on absences, justified and unjustified, as determined by the current government please refer to our [Bulletin article on attendance](#).

This in the database is Document 49:



| document_id INTEGER<br>PRIMARY KEY | link TEXT                                                                                       | source IN<br>REFERENCES<br>Source(sour |
|------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------|
| 49                                 | <a href="https://www.burnside.school.nz/absences/">https://www.burnside.school.nz/absences/</a> | 1 name: 'Bu                            |

Part of list of words connected to table 49:

| SELECT |        | *                   | FROM    | KeywordDocument                              | Schema      | Query Editor                                  | Auto Reload | Find | Other Tools... |
|--------|--------|---------------------|---------|----------------------------------------------|-------------|-----------------------------------------------|-------------|------|----------------|
|        | id     | INTEGER PRIMARY KEY | word_id | INT NOT NULL REFERENCES Keyword(word_id) idx | document_id | INT NOT NULL REFERENCES Document(document_id) | frequency   | INT  | +              |
| 87     | 100322 |                     | 212     | word: 'we' frequency:1380                    | 49          | link: 'https://...                            | 5           |      |                |
| 88     | 100323 |                     | 543     | word: 'offer' frequency:354                  | 49          | link: 'https://...                            | 1           |      |                |
| 89     | 100324 |                     | 272     | word: 'home' frequency:1454                  | 49          | link: 'https://...                            | 2           |      |                |
| 90     | 100325 |                     | 311     | word: 'if' frequency:1249                    | 49          | link: 'https://...                            | 5           |      |                |
| 91     | 100326 |                     | 226     | word: 'your' frequency:3301                  | 49          | link: 'https://...                            | 9           |      |                |
| 92     | 100327 |                     | 1277    | word: 'child' frequency:237                  | 49          | link: 'https://...                            | 9           |      |                |
| 93     | 100328 |                     | 219     | word: 'is' frequency:4624                    | 49          | link: 'https://...                            | 6           |      |                |
| 94     | 100329 |                     | 478     | word: 'go' frequency:467                     | 49          | link: 'https://...                            | 2           |      |                |
| 95     | 100330 |                     | 286     | word: 'be' frequency:3869                    | 49          | link: 'https://...                            | 4           |      |                |
| 96     | 100331 |                     | 2199    | word: 'absent' frequency:16                  | 49          | link: 'https://...                            | 5           |      |                |
| 97     | 100332 |                     | 310     | word: 'ani' frequency:756                    | 49          | link: 'https://...                            | 2           |      |                |
| 98     | 100333 |                     | 390     | word: 'reason' frequency:273                 | 49          | link: 'https://...                            | 4           |      |                |
| 99     | 100334 |                     | 218     | word: 'it' frequency:2821                    | 49          | link: 'https://...                            | 2           |      |                |
| 100    | 100335 |                     | 2654    | word: 'essenti' frequency:66                 | 49          | link: 'https://...                            | 1           |      |                |
| 101    | 100336 |                     | 221     | word: 'that' frequency:2787                  | 49          | link: 'https://...                            | 4           |      |                |
| 102    | 100337 |                     | 213     | word: 'you' frequency:5067                   | 49          | link: 'https://...                            | 7           |      |                |
| 103    | 100338 |                     | 561     | word: 'let' frequency:64                     | 49          | link: 'https://...                            | 1           |      |                |
| 104    | 100339 |                     | 277     | word: 'know' frequency:197                   | 49          | link: 'https://...                            | 1           |      |                |
| 105    | 100340 |                     | 214     | word: 'as' frequency:1851                    | 49          | link: 'https://...                            | 3           |      |                |
| 106    | 100341 |                     | 369     | word: 'soon' frequency:33                    | 49          | link: 'https://...                            | 1           |      |                |
| 107    | 100342 |                     | 292     | word: 'possibl' frequency:177                | 49          | link: 'https://...                            | 1           |      |                |
| 108    | 100343 |                     | 220     | word: 'import' frequency:386                 | 49          | link: 'https://...                            | 1           |      |                |
| 109    | 100344 |                     | 7886    | word: 'phonetext' frequency:1                | 49          | link: 'https://...                            | 1           |      |                |
| 110    | 100345 |                     | 250     | word: 'or' frequency:3508                    | 49          | link: 'https://...                            | 4           |      |                |
| 111    | 100346 |                     | 208     | word: 'email' frequency:155                  | 49          | link: 'https://...                            | 3           |      |                |
| 112    | 100347 |                     | 1734    | word: 'confirm' frequency:46                 | 49          | link: 'https://...                            | 1           |      |                |
| 113    | 100348 |                     | 263     | word: 'provid' frequency:837                 | 49          | link: 'https://...                            | 2           |      |                |
| 114    | 100349 |                     | 199     | word: 'with' frequency:2754                  | 49          | link: 'https://...                            | 2           |      |                |
| 115    | 100350 |                     | 1595    | word: 'acknowledg' frequency:34              | 49          | link: 'https://...                            | 1           |      |                |
| 116    | 100351 |                     | 252     | word: 'there' frequency:619                  | 49          | link: 'https://...                            | 1           |      |                |
| 117    | 100352 |                     | 261     | word: 'are' frequency:3103                   | 49          | link: 'https://...                            | 2           |      |                |
| 118    | 100353 |                     | 7760    | word: 'genuin' frequency:7                   | 49          | link: 'https://...                            | 1           |      |                |
| 119    | 100354 |                     | 236     | word: 'may' frequency:922                    | 49          | link: 'https://...                            | 1           |      |                |

The first paragraph, "If your child is going to be absent for any reason it is essential that you let us know as soon as possible." can be seen from id 100325 to id 100342. Further paragraphs are also in the list. "absence" is present with a frequency of 57, indicating that it is weighted heavily due to being in the title.